Airbnb Price Prediction: A Statistical and Machine Learning Approach

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By

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CERTIFICATE

I hereby certify that the project " **Airbnb Price Prediction: A Statistical and Machine Learning Approach**" is an authentic report of the project work carried out by Ms. PARVATHI M J, MSc. Applied Statistics and Data Analytics student in the Department of Statistics, University of Kerala, Kariavattom Campus, during 2023-2025 under my supervision and guidance, in partial fulfilment of the requirements for M.Sc. Degree in Applied Statistics and Data Analytics of the University of Kerala.

Kariavattom Dr. Manoj Chacko

July 2025 Supervising Guide

DECLARATION

I hereby declare that the project entitled " **Airbnb Price Prediction: A Statistical and Machine Learning Approach** " submitted to the University of Kerala in partial fulfilment of the requirements for awarding the Degree of Master in Applied Statistics and Data Analytics is a record of original project work done by me during the period of my study in Department of Statistics, University of Kerala, Kariavattom, Trivandrum.

Kariavattom PARVATHI M J

July 2025

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ABSTRACT

Airbnb Price Prediction: A Statistical and Machine Learning Approach explores Airbnb listings pricing dynamics in Paris using a robust integration of statistical analysis and machine learning methods.

The research starts with thorough data cleaning and preprocessing, then proceeds with exploratory data analysis to identify patterns, distributions, and outliers. Multiple statistical tests of hypothesis, such as Welch's t-test, one-way ANOVA, and the Kruskal-Wallis H-test, are employed to determine significance of price variation among listing types, availability statuses, and other categorizations. Geospatial K-Means clustering is utilized to harvest location-based price differences. Feature engineering and dimensionality reduction strategies like SelectKBest, Random Forest feature importance, and Principal Component Analysis (PCA) are used to improve efficiency of the model and mitigate overfitting.

A few supervised machine learning algorithms such as Linear Regression, Decision Tree, Random Forest, XGBoost, and LightGBM are trained to estimate log-transformed prices. All these models are assessed based on metrics such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared (R^2), with RandomizedSearchCV being used to tune hyperparameters for enhancing accuracy.

Finally, a real-time price prediction based on user-specified features is facilitated by an interactive interface developed with ipywidgets. This project proves the success of the integration of statistical inference with predictive modelling in offering both interpretability and useful tools for dynamic pricing within the Airbnb marketplace.

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CHAPTER 1
INTRODUCTION

1.1 INTRODUCTION

A large share of growth in the sharing or platform economy is driven by peer-to-peer (P2P) accommodation providers such as Airbnb. Airbnb and the broader sharing economy have had a significant impact not only on tourism and hospitality research but also on destination economies, the experience of place for both residents and guests, and the markets they serve. The emergence of globally dominant platform-based intermediaries — such as Airbnb, Booking.com, and Uber — has fundamentally transformed interactions and relationships between actors within the tourism ecosystem. Among these, Airbnb is often regarded as having one of the most disruptive influences on the accommodation sector and local destinations.



Figure 1: Airbnb Logo

Since its inception in 2008, Airbnb has been at the very heart of this revolution, offering a platform by which individuals can lease out houses, spare bedrooms, or unusual spaces (e.g., treehouses, yurts). This has opened up new possibilities for travellers and hosts alike, transforming how people travel and experience locations. Airbnb brings homeowners and guests worldwide together, offering unique spaces and experiences without ever taking ownership of any of the properties in the inventory. Instead, it offers the tools, protections, and assistance both groups require to rent and reserve their homes safely. Airbnb's platform has enabled the creation of employment opportunities across over 220 countries, facilitating various services such as cleaning, property management, maintenance, and tourism operations.

To hosts, it provides adjustable streams of income — from letting a single room on an irregular basis to running multiple properties professionally. The platform also facilitates pricing, booking management, guest messaging, and tracking earnings, as well as host protection insurance and integrated support services. From the perspective of a guest, Airbnb offers flexibility and choice, enabling one to stay in a number of different types of accommodations — city apartments, country cottages, and even non-conventional locations. Reviews, confirmed profiles, and 24/7 customer service have fostered trust and security for transactions.

Additionally, Airbnb's move into Experiences allows hosts to build special experiences, making the travel experience richer while offering new sources of revenue. Overall, Airbnb has not only revolutionized the hospitality sector but also fostered demand for local businesses and services, enabling owners of properties to let out undeveloped space and stimulating local economies in new and exciting ways.

1.2 RESEARCH PROBLEM

Although extensive data on Airbnb listings, calendars, and reviews is publicly available, many hosts continue to face challenges in setting optimal prices and understanding demand fluctuations. The dynamic nature of short-term rental markets, influenced by factors such as availability status, temporal patterns, listing attributes, and location, makes price optimization a complex task.

There is a clear need for a systematic and data-driven approach to analyze how these factors impact listing prices and availability. In addition, while machine learning models offer the potential for accurate price prediction, their application in this context — combined with robust feature selection and model interpretability — remains underexplored. Similarly, understanding trends in customer reviews over time can provide valuable insights for improving service quality and guest satisfaction, but such analyses are often overlooked.

This study seeks to address these gaps by applying comprehensive exploratory data analysis, inferential statistical testing, geospatial clustering, and predictive modelling techniques to Airbnb calendar, listings, and reviews datasets. The aim is to generate actionable insights that can assist hosts in making informed pricing decisions and understanding key patterns in customer engagement.

1.3 OBJECTIVES OF THE STUDY

The primary objectives of this study are to develop a comprehensive understanding of Airbnb pricing in Paris and to build a predictive framework using statistical and machine learning techniques. Specifically, the study aims to:

1. Examine the distribution of Airbnb listing prices, availability trends, and review trends to determine data characteristics and outliers.
2. Compare price distributions between groups (e.g., available vs. not available days, weekdays vs. weekends) using both graphical techniques and statistical hypothesis tests.
3. Determine price differences by availability status, day types, listing types, and locations.
4. Perform geospatial clustering to segment listings based on geographic coordinates and incorporate these clusters as predictive variables.
5. Select key features and reduce dimensionality through methods like SelectKBest, Random Forest feature importance, and Principal Component Analysis (PCA) to improve model performance and interpretability.
6. Build, evaluate, and compare regression models, including Linear Regression, Decision Tree, Random Forest, XGBoost, and LightGBM, for predicting log-transformed Airbnb prices.
7. Apply hyperparameter tuning and performance metrics (MSE, RMSE, MAE, R^2, and MAPE) to identify the most accurate and robust predictive models.

8. Develop an interactive price estimate interface with ipywidgets and Google Colab Notebook that allows users to input custom listing characteristics and receive live price estimates.

9. Explain feature importance of leading models and statistical testing to determine the most significant variables influencing Airbnb prices.

1.4 SCOPE AND SIGNIFICANCE OF THE STUDY

This study focuses on uncovering pricing patterns and demand trends along with key determinants of price via the analysis of the Airbnb data from calendar listings and reviews datasets.

- Price distributions, availability trends, and attribute relationships are understood through Exploratory data analysis (EDA).
- Meaningful price differences across categories, such as availability status plus weekends versus weekdays plus listing types, are determined using inferential statistics. Geographic groupings are found via clustering analysis. This analysis relies upon listing locations.
- Predictive models are built by machine learning algorithms for price estimation. They use too multiple feature engineering techniques.
- Actionable perceptions let hosts optimize prices as well as set competitive prices.
- Planned decision-making is helped as the comprehension of demand patterns is improved by it.
- Show the utility of machine learning in the sharing economy for predictive pricing.
- Add research on data-driven methods to platform economy analytics literature.

1.5 SOURCE OF DATA

The data used within this study comes from that publicly available Inside Airbnb dataset, being a project providing detailed datasets for various cities right across the globe. Analysed datasets mainly include:

- listings.csv has thorough data for each listing. Attributes like room type, location, host characteristics, and price are included.
- calendar.csv: For each listing, it contains daily price with availability data.
- reviews.csv: This contains visitor textual reviews, as well as reviewer details plus timestamps.

These datasets are appropriate for exploratory analysis in addition to predictive modelling; they offer a thorough view concerning the Airbnb ecosystem.

1.6 STRUCTURE OF THE REPORT

The report is organized into the following chapters:

- Chapter 1: Introduction. This chapter introduces the study, defines the research problem, lists the objectives and scope, and outlines the structure of the report.
- Chapter 2: Literature Review. This chapter summarizes relevant academic work on Airbnb pricing, machine learning models in the sharing economy, and factors that affect rental prices.

- Chapter 3: Research Methodology. This chapter details the step-by-step strategy adopted. It includes data acquisition, cleaning, feature engineering, statistical testing, clustering, model building, evaluation, and developing an interactive interface.
- Chapter 4: Data Description and Preparation. This chapter describes the data sources, cleaning processes, encoding, outlier treatment, feature creation, and how calendar and review data were used.
- Chapter 5: Data Analysis and Results. This chapter presents exploratory and inferential statistical analysis, clustering results, and findings from machine learning models. It includes a comparison of performance and feature importance.
- Chapter 6: Predictive Modelling. This chapter focuses on data splitting, scaling, dimensionality reduction (PCA), feature selection, model training, tuning, and the interactive price prediction tool.
- Chapter 7: Conclusions and Recommendations. This chapter summarizes key insights, research contributions, limitations, practical recommendations, and suggestions for future work.

CHAPTER 2
LITERATURE REVIEW

2.1 INTRODUCTION

The rise of Airbnb and other short-term rental platforms has greatly affected the global hospitality industry. The wealth of user-generated data from Airbnb listings, which includes pricing, reviews, amenities, and user behaviour, has opened doors for researchers and practitioners. They can now use machine learning and data science to understand pricing methods and guest preferences better. This chapter examines current literature on Airbnb price prediction, data analysis, and platform dynamics, focusing on methods, tools, and gaps in existing knowledge.

2.2 MACHINE LEARNING APPROACHES FOR AIRBNB PRICE PREDICTION

The use of machine learning (ML) for predicting Airbnb prices has received a lot of interest in recent years. This is largely because the platform offers rich and varied datasets. Researchers have looked into different modelling techniques to predict listing prices accurately. They use both structured and unstructured data, which includes numerical features, images, and text reviews.

2.2.1 Survey of Price Prediction Models

Prajapati and Suthar (2022) presented an in-depth survey that analysed various machine learning algorithms applied in predicting Airbnb listing prices. They were interested in learning the merits and demerits of different algorithms, such as linear regression, support vector machines (SVM), random forest, XGBoost, and neural networks. The research established that ensemble algorithms such as Random Forest and XGBoost generally performed better than basic methods in being more accurate and generalized.

In addition, the authors highlighted the significance of preprocessing steps like dealing with missing values, encoding categorical features, and normalizing numerical attributes. Their work also indicated that the performance of the models could differ a lot depending on feature selection strategies and hyperparameter tuning methods. This research provided the basis for understanding which models are best fit for Airbnb pricing issues based on data complexity as well as modelling objectives.

2.2.2 Predicting Prices Using Multiple Feature Modalities

Ahuja et al. (2021) took a new approach by suggesting a multi-modal learning framework to improve Airbnb price prediction. Instead of just using tabular data like room type, number of reviews, and location, their model added three more types of features: geolocation data, images of listings, and textual descriptions. This complete use of both structured and unstructured data allowed them to find deeper insights into the factors that influence price.

Their findings showed that combining different types of data greatly improved prediction accuracy. For example, listing images helped the model assess the visual quality of the property, while textual descriptions provided details about amenities, neighbourhood appeal, and host tone. These are aspects that structured features alone might miss. The study showed how deep

learning and feature fusion techniques can improve Airbnb price modelling in real-life situations.

2.2.3 Integrating Sentiment Analysis with Machine Learning

Rezazadeh Kalehbasti et al. (2021) provided an innovative solution by incorporating user review sentiment analysis into the price forecasting model. They employed Natural Language Processing (NLP) methods to derive sentiment scores from guest reviews and utilized these as supplementary features for training machine learning models. Their reason was that guest satisfaction, as indicated by review tone, could be a soft signal for property price.

This model worked well in capturing intangible features like trustworthiness of host, comfort of stay, or cleanliness—features frequently left out of structured data. Sentiment score integration enhanced model performance, and their findings emphasized the value of incorporating qualitative user feedback into predictive modelling.

2.3 USER BEHAVIOUR AND EXPERIENCE ANALYSIS

Some of the latest research started examining how users engage with Airbnb's platform, going beyond the properties' attributes to investigating the guest behaviour and decision-making. Jain et al. (2024) examined user browsing and booking logs to visualize repetitive browsing and booking behaviours. Their research employed visualization tools such as Sankey diagrams to monitor how users browse listings, apply filters, and make final booking decisions. They discovered that users tend to come back to listings several times, initially filtering based on price and location, and then by amenities and reviews. The study emphasizes the importance of quality images and positive guest feedback in influencing booking decisions. It also provides useful insights for improving Airbnb's user interface and recommendation systems. By shifting focus to the user's digital journey, this research highlights how user behaviour data can support better platform design and listing strategies.

2.4 EMERGING TRENDS AND SYSTEMATIC REVIEWS

Over the past few years, Airbnb studies have expanded beyond existing streams to cover emerging areas like amenity effects, user satisfaction, sustainability, and research methodology developments. This section showcases major contributions that capture the evolving research interests of academia and practice.

2.4.1 Impact of Amenities on Price and Satisfaction

Kumar et al. (2024) set out to explore the impact certain amenities have on both the price and perceived value of Airbnb listings. Employing data mining methods, they established that amenities such as air conditioning, Wi-Fi, free parking, and pet-friendliness are highly correlated with increased listing prices and enhanced guest reviews. The research indicates that visitors will pay a premium for convenience-related amenities, and hosts may strategically introduce high-impact amenities to enhance occupancy and ratings. This paper adds to the increased interest in the examination of how micro-level listing features impact macro-level performance measures.

2.4.2 Topic Modelling of Airbnb Research Evolution (2015–2022)

In a companion study, Sahu et al. (2023) performed topic modelling on a vast corpus of Airbnb studies between 2015 and 2022. The analysis revealed dominant trends and changes—

demonstrating early research concentrated on economic effects and urban-level effects, but later studies are focused on real-time analytics, user behaviour modelling, and machine learning. The trend of the topic emphasizes a shift from descriptive research to predictive and prescriptive analytics, consistent with wider shifts in the technology of tourism and digital marketplaces.

2.5 RESEARCH GAPS

While there have been multiple studies which furthered the research on Airbnb price forecasting and user behaviour, there are still significant gaps. One of the limitations is the absence of geographical diversity in the data sets used; most of the work is focused on listings in large cities such as New York, San Francisco, or Barcelona, making the results less generalizable to smaller cities or new markets. Another significant gap is the restricted use of explainable AI methods. While intricate models like neural networks and ensemble algorithms are frequently used, not many studies make efforts to interpret or visualize their predictions, and it becomes challenging for hosts, stakeholders, or policymakers to comprehend the model's reasoning. Furthermore, there is an evident lack of utilization of real-time behavioural information, e.g., user navigation patterns or booking hours, that could more effectively improve the accuracy of predictions and responsiveness on platforms. Lastly, the majority of studies consider Airbnb prices as fixed and do not account for dynamic pricing strategies responsive to demand, seasonality, or local events—a common application in the hotel sector yet not yet fully explored in peer-to-peer accommodation studies.

2.6 SUMMARY

This review of the literature discussed major advancements in Airbnb studies in the domains of machine learning applications, user experience research, and new trends. It emphasized how complex modelling methods, feature-laden data, and behavioural analysis are designing the future of Airbnb prices. The reviews considered here affirm the trend of the current project, particularly in the use of machine learning on both structured and unstructured Airbnb data. But in pointing out where the gaps lie—specifically in explainability, geographical representation, and dynamic modelling—this project stands poised to add significant value to Airbnb price prediction models.

CHAPTER 3
RESEARCH METHODOLOGY

3.1 OVERVIEW OF THE RESEARCH DESIGN

This study uses a data-based method to explore and predict pricing trends in Airbnb listings with real-world data from Paris. The approach includes:

- Exploratory Data Analysis (EDA)
- Statistical testing
- Geospatial clustering
- Feature selection
- Supervised machine learning models
- Hyperparameter tuning
- Interactive interface design

The goal is to provide useful insights into the factors affecting Airbnb prices and to create a tool for estimating dynamic prices.

3.2 EXPLORATORY DATA ANALYSIS (EDA)

EDA techniques were used to find patterns, trends, and outliers in the dataset:

- Univariate Analysis:

  - Histogram: A bar plot that shows the frequency distribution of a single variable. It is helpful for visualizing the distribution, skewness, and spread. This technique was applied to variables like price and number of reviews.
  - Boxplot: Displays the five-number summary, which includes minimum, first quartile, median, third quartile, and maximum. Outliers appear as points. This method helps identify extreme values.
  - Boxen Plot: A more advanced version of the boxplot that is especially good at showing distributions with heavy tails or large datasets. It focuses on the distribution of values in the tails and helps detect patterns not visible in a standard boxplot.

- Bivariate and Multivariate Analysis:

  - Violin Plot: This combines a boxplot with a kernel density plot. It shows both the shape of the distribution and summary statistics across groups. A violin plot consists of four components. A white Centred Dot at the middle of the graph - The white dot point at the middle is the median of the distribution. A thin grey bar inside the plot - The bar in the plot represents the Quartile range of the distribution. A long thin line coming outside from the bar - The thin line represents the rest of the distribution which is calculated by the formulae Q1-1.5 IQR for the lower range and Q3+1.5 IQR for the upper range. The point lying beyond this line are considered as outliers. A line boundary separating the plot- A KDE plot is used for defining the boundary of the violin plot it represents the distribution of data points.
  - Heatmap: A color-coded correlation matrix. It is effective for spotting strong linear relationships or collinearity between numeric variables.

- Temporal Patterns:

  - Line Plots: These are used to show trends over time. They help visualize review trends and availability changes across months or days of the week.

- Correlation Analysis:

  - Pearson Correlation: This measures the strength and direction of linear relationships between continuous variables, such as price and number of accommodations.
  - Spearman Correlation: Correlation measures the strength of the association between two continuous variables and is useful when data contains outliers or is not normally distributed.

3.3 STATISTICAL TESTING

Inferential statistics were used to test assumptions and validate group differences.

- **Welch's t-test:**

  Welch's t-test is used to compare the means of two independent groups when the assumption of equal variances is not met. In this study, it was applied to compare average prices between weekend and weekday listings, as well as between available and unavailable listings. It is regarded as the parametric equivalent of the Two-Sample T-test.

$$t = \frac{\overline{X_1} - \overline{X_2}}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

  where:
  $\bar{X}_1$, $\bar{X}_2$ = sample means
  s_1^2, s_2^2 = sample variances
  n_1, n_2 = sample sizes
  The degrees of freedom are calculated using the Welch–Satterthwaite equation:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

  Assumptions:
  - Samples are independent
  - Data is approximately normally distributed
  - Variances are unequal

- **One-way ANOVA:**

  Statistical test used to determine whether there are statistically significant differences between the means of three or more independent groups.

$$F = \frac{MS_{between}}{MS_{within}}$$

  where:

$MS_{between}$ is the mean square between groups (variability due to differences between group means)
MS_{within} is the mean square within groups (variability within each group)

Assumptions checked before applying ANOVA:

1. Shapiro-Wilk Test:
   This test checks whether the residuals (errors) from the model are normally distributed.
   - Null Hypothesis (H_0): Residuals are normally distributed.
   - Alternative Hypothesis (H_1): Residuals are not normally distributed.
   - If $p > 0.05$: Fail to reject H_0 → data is approximately normal
   - If $p \leq 0.05$: Reject H_0 → data deviates significantly from normality

2. Levene's Test:
   This test evaluates whether the variances of the groups being compared are equal.
   - Null Hypothesis (H_0): Variances are equal across groups.
   - Alternative Hypothesis (H_1): At least one group has a significantly different variance.
   - If $p > 0.05$: Homogeneity of variances assumed
   - If $p \leq 0.05$: Variances are unequal → consider alternative tests

- **OLS Regression with Robust Standard Errors (HC3):**

  Ordinary Least Squares (OLS) regression assumes constant variance of residuals (homoscedasticity). When this assumption is violated, standard error estimates can become unreliable, leading to incorrect p-values. To address this, the HC3 variant of robust standard errors was used. HC3 adjusts the standard errors to account for heteroscedasticity, allowing for more valid statistical inference. In this study, it was applied to ensure reliable hypothesis testing in the presence of unequal variances.

- **Kruskal-Wallis H-test:**

  The Kruskal-Wallis H-test is a non-parametric alternative to one-way ANOVA, used when the assumptions of normality or homogeneity of variances are violated. It compares the distributions of three or more independent groups by testing for statistically significant differences in their median ranks.

$$H = \frac{12}{N(N+1)} \sum_{i=1}^{k} \frac{R_i^2}{n_i} - 3(N+1)$$

  where:

  - H = Kruskal-Wallis test statistic
  - N = total number of observations across all groups
  - k= number of groups
  - R_i^2 = sum of ranks in group i
  - n_i= number of observations in group i

- **Chi-square Test of Independence:**

The Chi-square test of independence is used to determine whether there is a significant association between two categorical variables. It compares the observed frequencies in a contingency table to the frequencies expected under the assumption of independence. A significant result indicates that the variables are not independent and may be statistically related. In this study, it was applied to explore associations between categorical features such as room type and availability.

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Where:

- χ^2= Chi-square statistic
- O_{ij} = observed frequency in cell i,j
- E_{ij}= expected frequency in cell i,j calculated as:

$$E_{ij} = \frac{(Row\ Total_i) \times (Column\ Total_j)}{Grand\ Total}$$

The test statistic follows a chi-square distribution with degrees of freedom df=(r−1)(c−1), where r is the number of rows and c is the number of columns.

3.4 SPATIAL CLUSTERING

To capture geographic influences:

- K-Means Clustering: This unsupervised learning algorithm is applied to latitude and longitude. It segments listings into spatial clusters.
- Elbow Method and Silhouette Score: The elbow method were evaluated to find the best number of clusters. The Silhouette Score evaluates how well each data point fits within its assigned cluster, and how distinctly separated it is from other clusters.
- The assigned cluster labels served as a new feature to show localized price patterns.

3.5 FEATURE SELECTION AND DIMENSIONALITY REDUCTION

- SelectKBest (f_regression): Chooses features that have the strongest linear connection to the target. This helps reduce noise and improve model performance.
- Random Forest Feature Importance: Measures how much each feature contributes to prediction by looking at impurity reduction.
- Principal Component Analysis (PCA): Transforms data to a lower-dimensional space by turning correlated variables into uncorrelated principal components. It retains 43 components that explain 95% of the total variance.

3.6 MACHINE LEARNING FRAMEWORK

3.6.1 Machine Learning

Machine learning is a part of artificial intelligence that focuses on creating systems that can learn from data and make decisions based on it. Rather than being programmed with specific rules, machine learning algorithms find patterns in data to carry out tasks like prediction, classification, or clustering.

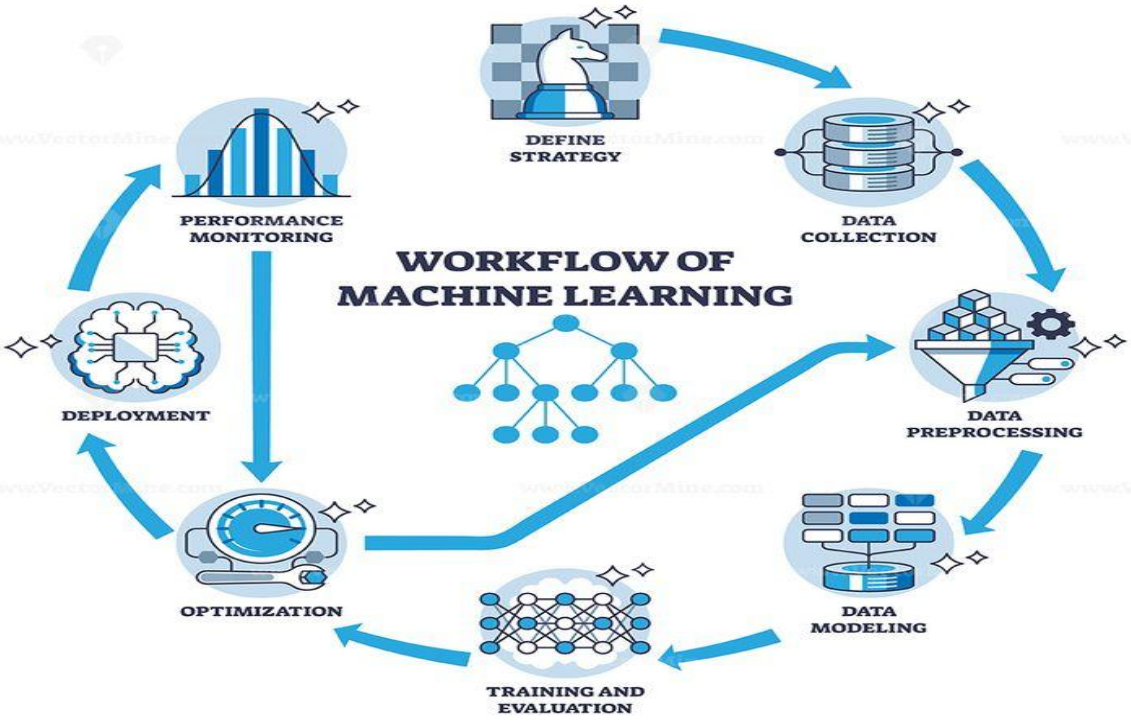


Figure 2: Workflow of ML

3.6.2 Types of Machine Learning

- Supervised Learning: In this type of learning, the algorithm trains on labelled data, which is in the form of input-output pairs. This project employed supervised learning to estimate price, the dependent variable or target we wanted to predict using features such as room type, location, and amenities offered.
- Unsupervised Learning: No labels are supplied with the data. An algorithm analyses patterns or groupings within that dataset. One example is K-Means clustering, which can also be used for segmentation based on geographical location.
- Semi-supervised Learning: This method works with a limited set of labelled data and a larger pool of unlabelled data. It was not utilized in this project.
- Reinforcement Learning: Algorithms derive insights by engaging with an environment where they're given rewards or penalties based on feedback gathered. This method was not applied in this project.

3.6.3 Regression in machine learning

Regression in machine learning refers to a supervised learning technique where the goal is to predict a continuous numerical value based on one or more independent features. Regression in machine learning is a fundamental technique for predicting continuous outcomes based on input features. It finds relationships between variables so that predictions can be made. we have two types of variables present in regression:

- Dependent Variable (Target): The variable we are trying to predict e.g. house price.
- Independent Variables (Features): The input variables that influence the prediction e.g. locality, number of rooms.

Applications of Regression

- Predicting prices: Used to predict the price of a house based on its size, location and other features.
- Forecasting trends: Model to forecast the sales of a product based on historical sales data.
- Identifying risk factors: Used to identify risk factors for heart patient based on patient medical data.
- Making decisions: It could be used to recommend which stock to buy based on market data.

Advantages of Regression

- Easy to understand and interpret.
- Robust to outliers.
- Can handle both linear relationships easily.

Disadvantages of Regression

- Assumes linearity.
- Sensitive to situation where two or more independent variables are highly correlated with each other i.e multicollinearity.
- May not be suitable for highly complex relationships.

3.6.4 Algorithms Used

- Linear Regression: A simple regression algorithm that assumes a straight-line relationship between features and the target. It serves as a performance standard.
- Decision Tree Regressor: A structure like a flowchart where each node splits the data based on the most important feature. It can overfit if not trimmed.
- Random Forest Regressor: A group of decision trees. Each tree helps decide the final output, which reduces overfitting and improves generalization.
- XGBoost Regressor: An optimized gradient boosting algorithm that builds models one after another, correcting the errors from earlier models. It is known for its high accuracy and speed.
- LightGBM Regressor: A gradient boosting model that uses leaf-wise growth and histogram-based methods for efficient computing.

3.6.5 Python Libraries and Tools

- pandas, numpy: For handling structured data and performing numerical operations.
- matplotlib, seaborn: For making static, animated, and interactive visualizations.
- scikit-learn: Provides tools for preprocessing, selecting models, training, and evaluation.
- xgboost, lightgbm: Specialized libraries for gradient boosting models.
- statsmodels, scipy.stats: Offers functions for statistical tests and regression modeling.
- ipywidgets: Allows interactive widgets in Google Colab Notebooks, used for the dynamic prediction interface.

3.7 MODEL EVALUATION

The following methods were used to assess model performance:

- Mean Squared Error (MSE): assesses the average squared variation between projected and real values. Large errors are sensitive.

- Root Mean Squared Error (RMSE): The square root of MSE, root mean squared error (RMSE) may be understood in the same units as the target variable.
- Mean Absolute Deviation (MAE): This is the average of the absolute differences between predicted and actual values.
- R^2 score: This represents the degree of variation in the target explained by the model.
- Mean Absolute Percentage Error (MAPE): Expressing error as a percentage, MAPE is helpful for comparing across several scales.

3.8 HYPERPARAMETER TUNING

Hyperparameter tuning optimizes model parameters that aren't learned from the data but are set before training. These parameters include the learning rate, maximum tree depth, number of estimators, and subsample ratios. Proper tuning greatly improves model performance and generalization.

- RandomizedSearchCV is a method that searches through a range of hyperparameter values using random sampling. It is more efficient than grid search.
- This method is used to optimize XGBoost and LightGBM models with cross-validation.

3.9 INTERACTIVE PRICE PREDICTION INTERFACE

An interactive interface was built with ipywidgets to allow dynamic pricing simulation.

- Users can enter features like the number of bedrooms and room type.
- The model updates price predictions in real-time.
- The predictions are transformed back from log scale to give clear results.

3.10 SUMMARY

This chapter explained the method used for the project. It covered each visualization, statistical technique, machine learning model, and software tool that was used. The combination of exploratory data analysis, statistical tests, machine learning, and an interactive system created a strong process for predicting Airbnb prices.

CHAPTER 4
DATA DESCRIPTION AND PREPARATION

4.1 DATA SOURCE

The data for this study comes from the publicly available Inside Airbnb dataset, which offers detailed datasets for cities worldwide. For this project, we chose data related to Paris because it is one of the most active short-term rental markets globally.

The primary datasets analysed include:

- listings.csv: This file contains detailed information on each listing, including room type, location, host characteristics, and price.
- calendar.csv: This file includes daily availability and price data for each listing.
- reviews.csv: This file has guest reviews along with timestamps and reviewer details.

These datasets provide a clear view of the Airbnb ecosystem and are useful for exploratory analysis and predictive modelling.

4.2 DATASET DESCRIPTION

The combined dataset contains 91,031 listings and a total of 74 variables from listings.csv. It is supported by supplementary data from calendar.csv and reviews.csv. Below is a categorized breakdown of the variables across all three datasets:

A. Identification

- id, scrape_id: Unique identifiers for listings and scraping sessions
- last_scraped, calendar_last_scraped: Date of data collection
- source: Source of scraping (typically “city scrape”)

B. Listing Content & Description

- name: Listing title
- description, neighborhood_overview: Unstructured textual data (not used in modeling)

C. Host Information

- host_id, host_name: Host identifiers
- host_since: Date the host joined Airbnb
- host_location, host_about: Host information
- host_response_time, host_response_rate, host_acceptance_rate: Host behavioral metrics
- host_is_superhost, host_has_profile_pic, host_identity_verified: Host status flags
- host_neighbourhood: Neighborhood of host
- host_listings_count, host_total_listings_count: Number of listings by host
- host_verifications: Methods of host verification

D. Location Details

- neighbourhood, neighbourhood_cleansed: Neighborhood information
- latitude, longitude: Geographical coordinates

E. Property Details

- property_type, room_type: Property category and room access type
- accommodates: Number of guests that can be hosted
- bathrooms, bathrooms_text: Number and description of bathrooms
- bedrooms, beds: Sleeping arrangement details
- amenities: List of amenities offered (text format)

F. Pricing and Availability

From listings.csv:

- price: Nightly price (target variable)
- minimum_nights, maximum_nights: Stay limits
- minimum_minimum_nights,maximum_minimum_nights, minimum_maximum_nights, maximum_maximum_nights: Constraints on booking durations
- minimum_nights_avg_ntm, maximum_nights_avg_ntm: Average minimum/maximum nights in the next 12 months
- has_availability: Availability flag
- availability_30, availability_60, availability_90, availability_365: Availability over 30, 60, 90, and 365 days

From calendar.csv:

- listing_id: Listing identifier
- date: Date of availability/pricing record
- available: Availability status (t/f)
- price: Price on that date
- adjusted_price: Adjusted nightly price (with discounts applied)
- minimum_nights, maximum_nights: Stay limits for that specific date

G. Review Data

From listings.csv:

- number_of_reviews, number_of_reviews_ltm, number_of_reviews_l30d: Total and recent reviews
- first_review, last_review: Dates of first and latest reviews
- review_scores_rating, review_scores_accuracy, review_scores_cleanliness, review_scores_checkin, review_scores_communication, review_scores_location, review_scores_value: Review sub-ratings
- reviews_per_month: Monthly review frequency

From reviews.csv:

- listing_id: Listing identifier
- id: Review ID
- date: Date of review
- reviewer_id: Unique reviewer ID
- reviewer_name: Name of reviewer

- comments: Text of review

H. Licensing and Booking

- license: Regulatory data (often incomplete)
- instant_bookable: Indicates instant booking availability

I. Calculated Host Listing Counts

- calculated_host_listings_count: Total listings managed by a host
- calculated_host_listings_count_entire_homes, calculated_host_listings_count_private_rooms, calculated_host_listings_count_shared_rooms: Room type–specific counts

4.3 DATA CLEANING AND PREPROCESSING STEPS

- Handling Missing Values:
- Numeric fields such as host_listings_count, minimum_nights, and maximum_nights were filled using the median.
- Review score columns were set to zero if there were no reviews.
- The host_since field was completed with the median year, using January 1st as a fallback.
- Bathrooms, bedrooms, and beds were filled based on mode or median, depending on the accommodates value.
- Categorical features received an "Unknown" label.
- In calendar.csv, minimum_nights and maximum_nights were filled with the median for each listing.
- In reviews.csv, rows with missing or empty comments or invalid dates were removed.
- Handling Data Types:
  - Date fields like host_since, calendar.date, and reviews.date were changed to datetime format.
  - Price columns were cleared of symbols and commas, then converted to float.
  - The available column in calendar.csv was changed to 1 or 0.

4.4 ENCODING

- Ordinal features, like host_response_time and stay_category_ntm, were label encoded.
- Binary features, such as host_is_superhost and host_identity_verified, were converted to 0 or 1.
- Nominal features, including room_type, property_type, and neighbourhood_cleansed, were processed using one-hot encoding.

4.5 OUTLIER HANDLING

- Outliers in minimum_nights, maximum_nights, host_listings_count, and host_total_listings_count were clipped using the IQR method.
- Avg_stay_ntm was clipped at the 99th percentile.
- Outliers in accommodates, bathrooms, bedrooms, and beds were kept, as they likely represent large properties.
- In calendar.csv, price outliers were capped at the 99th percentile.

4.6 FEATURE ENGINEERING

Several new features were created:

- month: Extracted from calendar.date for monthly trend analysis
- day_of_week: Indicates the day (0 = Monday to 6 = Sunday)
- is_weekend: Binary flag for weekends (Saturday or Sunday)
- year_month: Aggregated review period from reviews.date
- host_active_years: Time from host_since to last_scraped
- avg_min_nights: Average of minimum_minimum_nights and maximum_minimum_nights
- min_nights_category: Categorizes avg_min_nights into 'Short Stay', 'Week', 'Month', etc.
- avg_max_nights: Median of minimum_maximum_nights and maximum_maximum_nights
- max_nights_category: Groups avg_max_nights into logical stay lengths
- stay_category_ntm: Derived from avg_stay_ntm with categories like 'Short Stay', 'Long Stay', etc.

CHAPTER 5
DATA ANALYSIS AND RESULTS

5.1 DESCRIPTIVE STATISTICS

Descriptive statistics were calculated to understand the distribution of numerical features. This included measures like the mean, median, standard deviation, and quartiles.

Table 1: Descriptive Statistics of Key Features

Feature	Mean	Median	Std Dev	Min	Max
accommodates	3.33	3.00	1.75	1.00	16.00
bathrooms	1.19	1.00	0.53	0.00	42.00
bedrooms	1.29	1.00	0.90	0.00	33.00
beds	1.78	1.00	1.17	0.00	17.00
price	179.85	100.00	402.32	10.83	730.00
number_of_reviews	8.01	2.00	13.58	0.00	780.00
review_scores_rating	4.79	4.88	0.34	0.00	5.00

5.2 UNIVARIATE ANALYSIS

1. PRICE(TARGET):

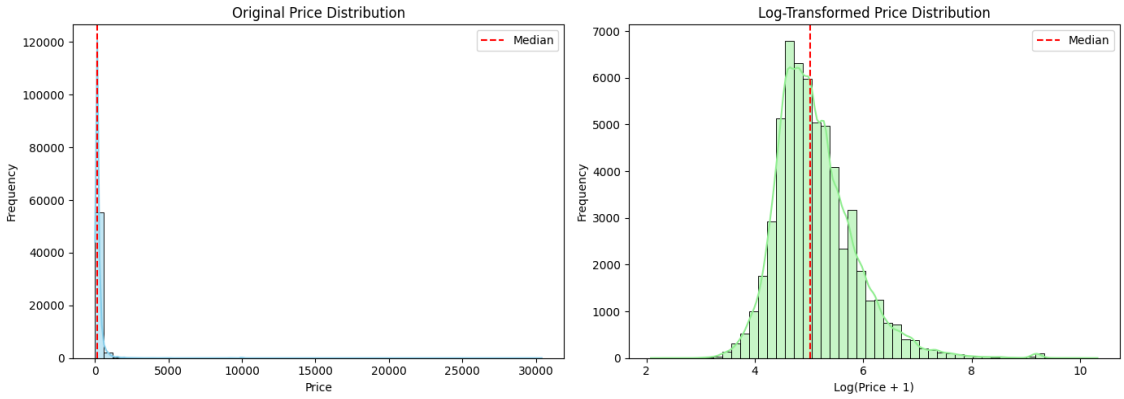


Figure 3: Price And Log Price

The original price distribution is highly skewed, indicating that most listings are at lower prices with a long tail of very expensive listings. The log transformation successfully reduces the skewness, making the distribution more symmetrical and suitable for linear models that assume normally distributed errors.

2. HOST SINCE YEAR DISTRIBUTION:

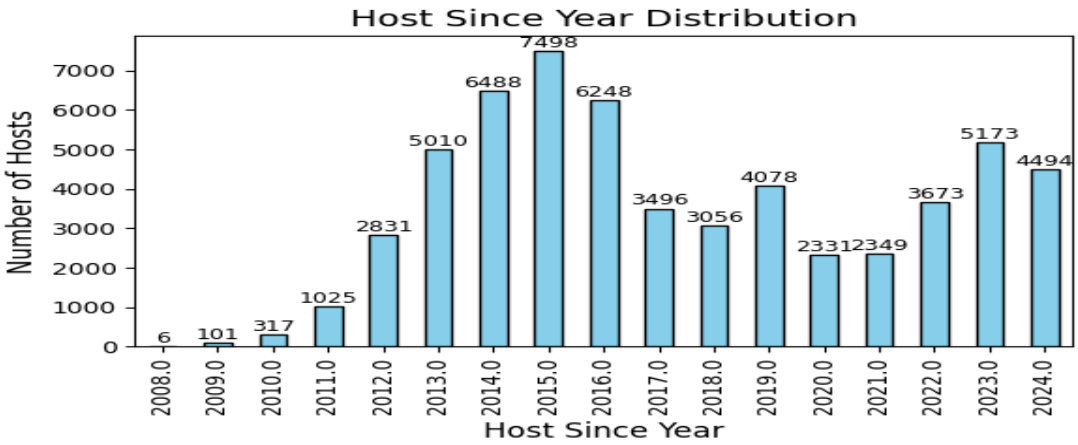


Figure 4: Host Since Year Distribution

This plot shows the number of hosts who joined Airbnb in different years. It provides insight into the growth of the platform and the age of the host community. A large number of recent hosts might indicate rapid market expansion, while a substantial number of older hosts suggests platform maturity and host loyalty. A decline of values in year 2020 suggest effect of CORONA period.

3. HOST ACTIVE YEARS DISTRIBUTION:

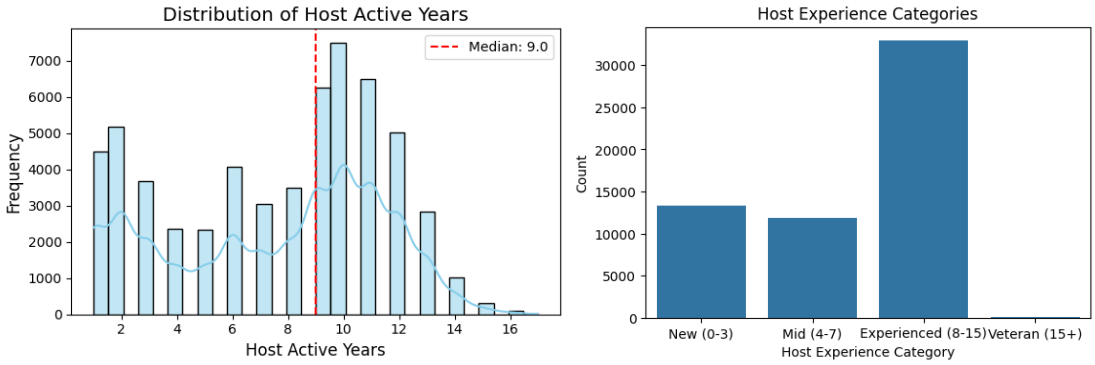


Figure 5: Host Active Years Distribution

This plot shows the distribution of how long hosts have been active on the platform. It helps understand the experience level of the host base. Knowing the typical host tenure can be useful for targeted host support or understanding market saturation.

Categorizing host activity into experience levels provides a clearer view of the proportion of new, mid-level, experienced, and veteran hosts. This can inform marketing strategies or host support programs.

4. KEY LISTING CHARACTERISTICS: HOST ENGAGEMENT, REGULATIONS, HOST QUALITY, AND BOOKING CONVENIENCE

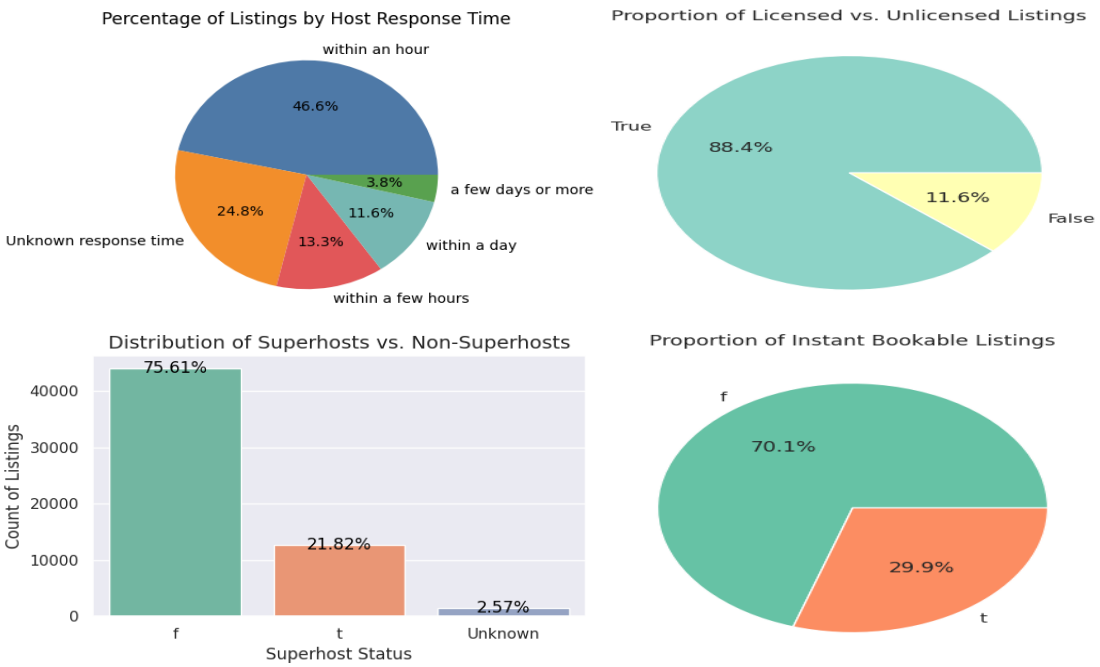


Figure 6.1: Host Response Time, Figure 6.2: Licensing Status

Figure 6.3: Super host Status, Figure 6.4: Instant Booking Status

Host Response Time: This indicates the percentage of listings by host response time to guest inquiries. Higher percentages in shorter response times tend to indicate greater host involvement and possibly enhanced guest experience, which could impact booking.

Licensing Status: This indicates the percentage of officially licensed listings. Licensing may indicate compliance with local law and possibly affect the legality and long-term sustainability of listings.

Super host Status: This indicates the percentage of listings by Super hosts versus normal hosts. Super host status is one of the main measures of host quality and performance according to parameters such as ratings, response rate, and cancellations.

Instant Booking Status: This indicates the percentage of listings where instant booking without host approval is allowed. Instant booking provides ease to guests and is preferred in many cases, which may affect visibility and bookings.

5. HOST ENGAGEMENT METRICS: RESPONSE AND ACCEPTANCE RATES:

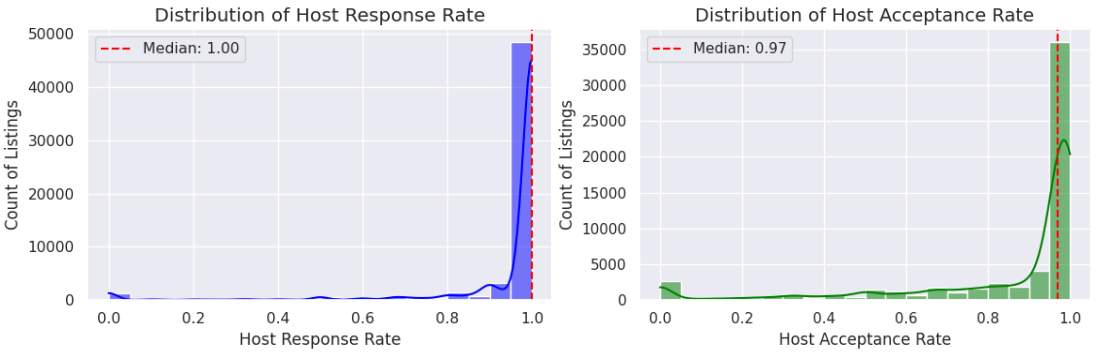


Figure 7.1: Host Response Rate, Figure 7.2: Host Acceptance Rate

HOST RESPONSE RATE: It indicates how often hosts reply to booking requests. Hosts who respond often are usually preferred as it implies responsive hosts and is typically associated with higher ratings on the platform.
HOST ACCEPTANCE RATE: This indicates the frequency with which hosts accept booking requests. An acceptance rate that is high reflects a willingness to host and is yet another popular metric used by platforms for ranking.

6. NEIGHBOURHOOD:

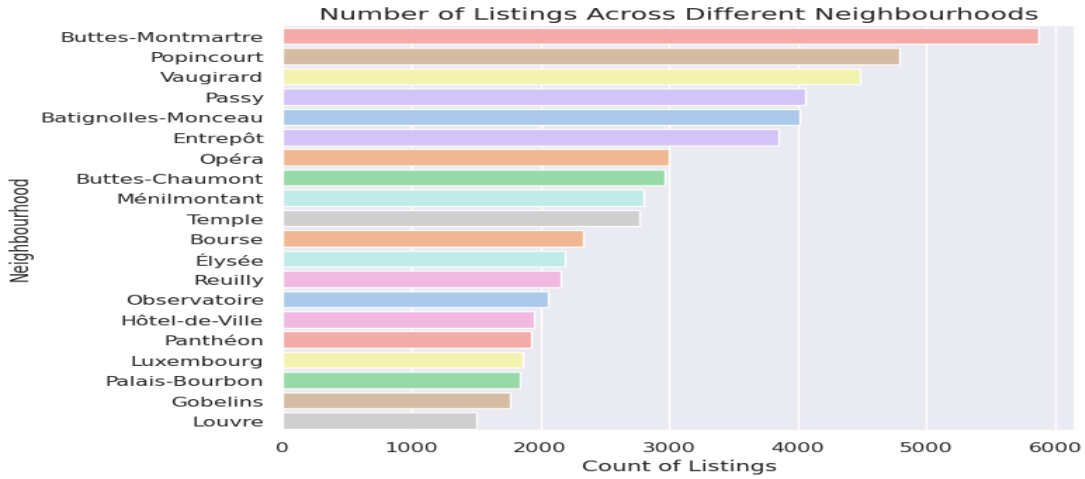


Figure 8: Neighbourhood

This plot shows the distribution of listings across different neighbourhoods in Paris. It highlights the most popular or saturated areas for Airbnb listings, which is crucial for understanding market concentration and potential demand/supply imbalances.

7. ACCOMMODATION TYPE BREAKDOWN: PROPERTY AND ROOM TYPES:

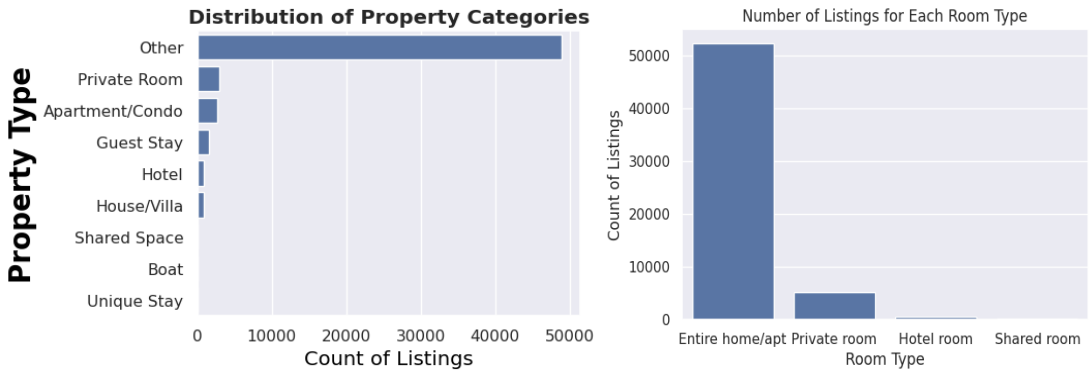


Figure 9.1: Property Categories, Figure 9.2: Room Types

PROPERTY TYPE DISTRIBUTION: This indicates the split of listings by categorized property types (e.g., Apartment/Condo, House/Villa). It indicates the most prevalent types of accommodation on offer, which can impact booking habits and target markets.
ROOM TYPE DISTRIBUTION: This graph illustrates the breakdown of listings by whether they provide a whole home/apartment, a private room, or a shared room. This is a core attribute of an Airbnb listing and affects privacy and guest experience.

8. TEMPORAL REVIEW TRENDS:



Figure 10: Yearly review volume

The "Monthly Review Volume" chart easily indicates an enormous surge in Airbnb reviews over time; from the time it was launched approximately in 2010 to the present time in 2024. This chart indicates that more and more individuals are utilizing Airbnb and leaving reviews, illustrating the robust growth of the platform. Although there was a significant fall off in early 2020 (presumably as a result of the pandemic), the number of reviews rebounded rapidly and continued to increase even more rapidly, reaching its peak in recent times. Having so much review data is extremely helpful in determining how many individuals use Airbnb, how quickly it has expanded, and just how significant events impact it.

5.3 MULTIVARIATE ANALYSIS

1. PRICE VS. HOSTING TENURE

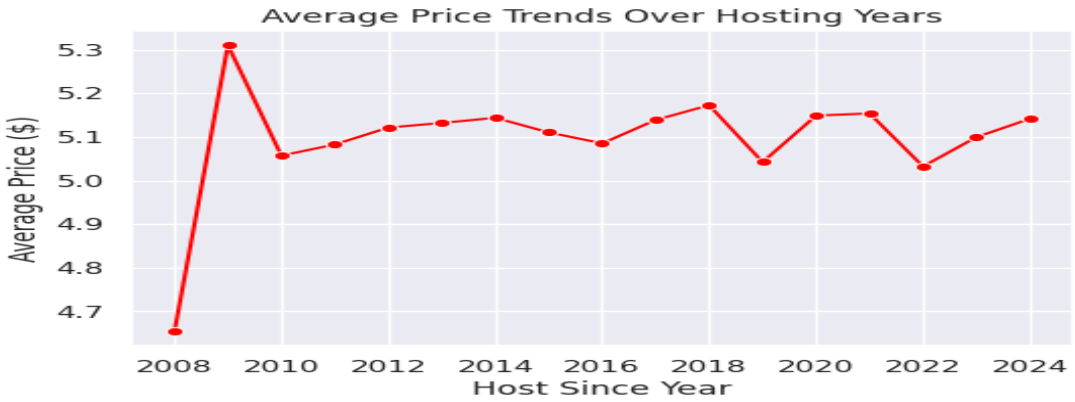


Figure 11: Price Vs. Hosting Tenure

The graph indicates that the mean listing price is not always related to how long a host has been on the site. Rather, it indicates fluctuations between different cohorts of hosts that joined in various years.

2. PRICE VS. HOST RESPONSE TIME:



Figure 12: Price Vs. Host Response Time

This plot is a comparison of the distribution of prices in terms of how fast hosts respond. There isn't a visible strong or clear correlation between the typical response time of a host and the price for their listings, according to this plot.

3. PRICE VS. NEIGHBOURHOOD:

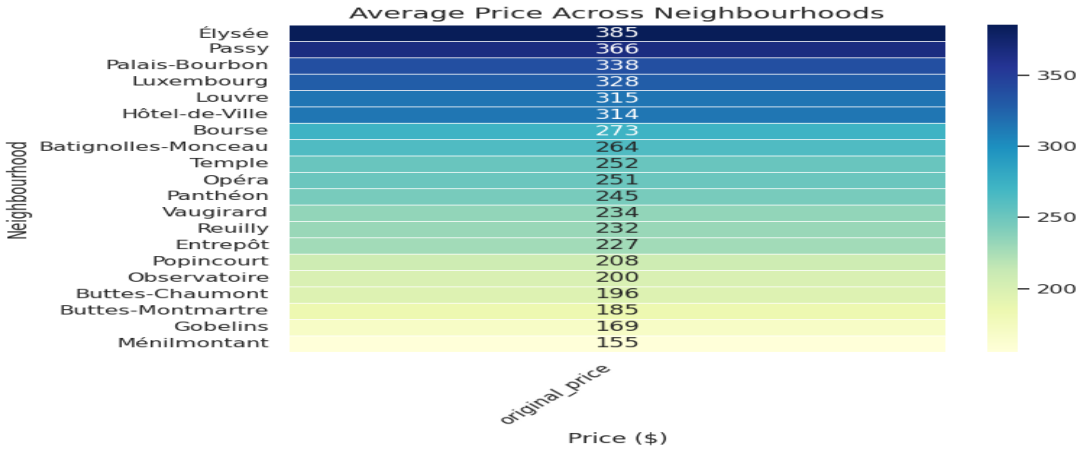


Figure 13: Price Vs. Neighbourhood

This heatmap visually identifies the large range in the average list prices throughout the various neighbourhoods. Certain neighbourhoods, for instance, Élysée and Passy, are far more costly on average compared to others, for example, Gobelins and Ménilmontant.

4. PRICE VS. PROPERTY TYPE

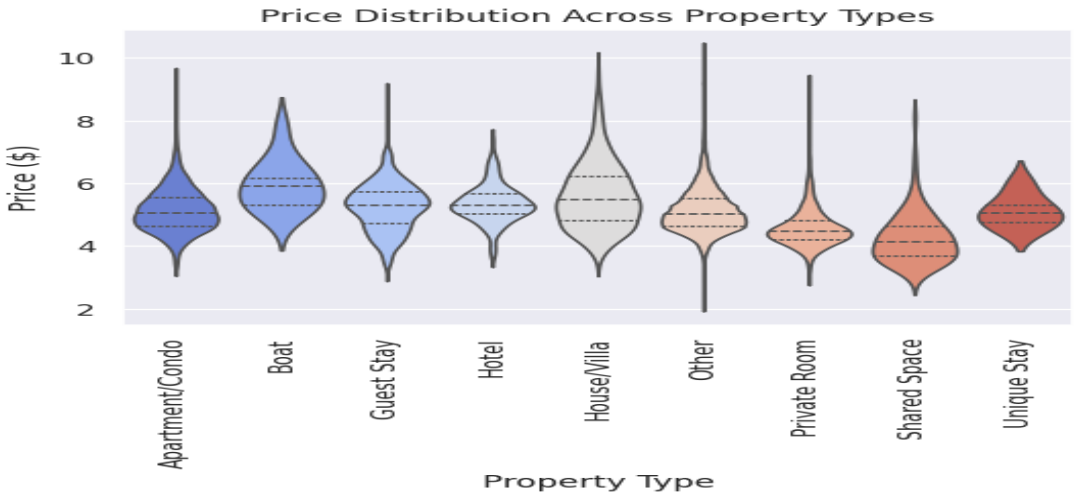


Figure 14: Price Vs. Property Type

This graph indicates that property type is an important factor affecting listing prices. Whole home/apartment types such as Houses/Villas are the costliest with shared spaces being the lowest priced. Private rooms are in the middle.

5. PRICE VS. ROOM TYPE



Figure 15: Price Vs. Room Type

This boxen plot well illustrates that listing price is largely driven by room type. Whole houses/apartments are the costliest, then hotel rooms, private rooms, and shared rooms are the least costly.

6. PRICE VS. HOST VERIFICATION STATUS

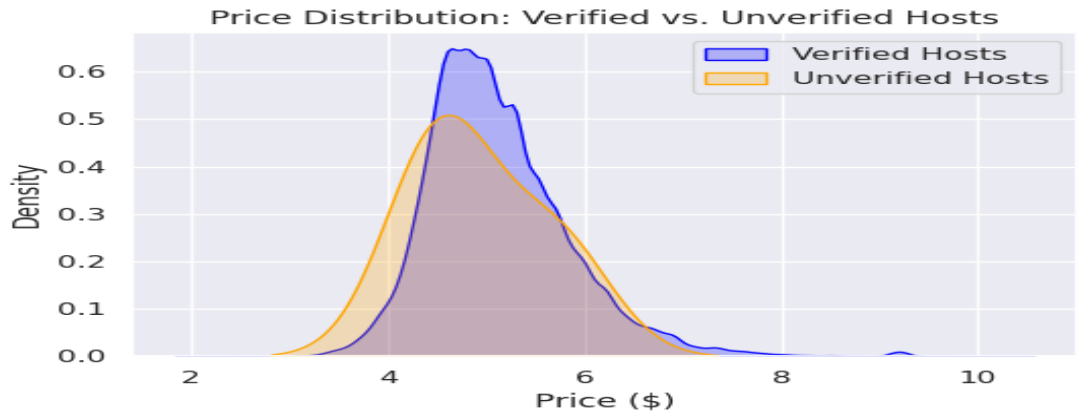


Figure 16: Price Vs. Host Verification Status

This KDE plot indicates that perhaps there is a slight variation in the pricing behaviour of verified and non-verified hosts. Verified hosts, at an average rate, charge slightly more for their listings than unverified hosts do. They also seem to have a greater percentage of listings in the higher tiers. This may be due to some number of reasons, including that guests may be more likely to pay a premium for verified host listings, or that verified hosts offer different types of or better-quality properties. But the main price concentration for both groups remains roughly the same.

7. PRICE VS. ACCOMMODATION CAPACITY

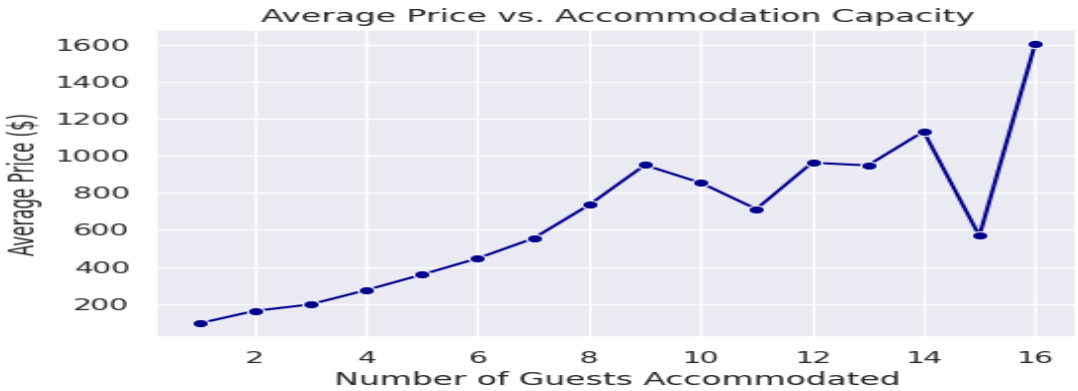
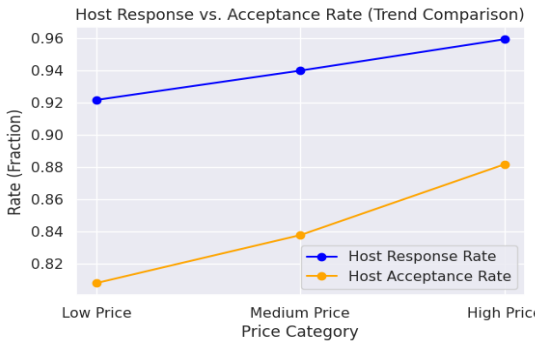


Figure 17: Price Vs Accommodation Capacity

The graph indicates a positive relationship between accommodation capacity and average price, which is logical given that the bigger the group, the bigger and costlier the properties usually are.

8. PRICE VS. HOST RESPONSE/ACCEPTANCE RATES



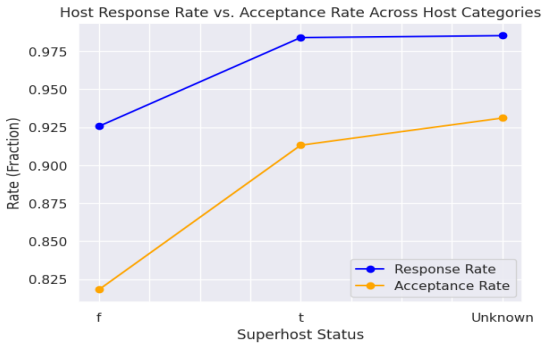


Figure 18.1: Price Vs. Host Response/Acceptance Rates

Figure 18.2: Super host Vs. Host Response/Acceptance Rates

This plot indicates a relationship between the price tier of a listing and host engagement metrics. Hosts of more expensive listings have both higher response rates and much higher acceptance rates than hosts of less expensive listings.

This plot overwhelmingly supports that Superhosts have much higher host engagement metrics than non-Superhosts. They reply to booking requests far more often and show acceptance rates many times higher as well. These actions comply with Superhost requirements, which involve holding a strong response and acceptance rate. The "Unknown" category is an indication that listings with no known Superhost status may not only fall between normal hosts and Superhosts who have received the Superhost badge but may have engagement rates in between as well.

9. PRICE VS. AVAILABILITY

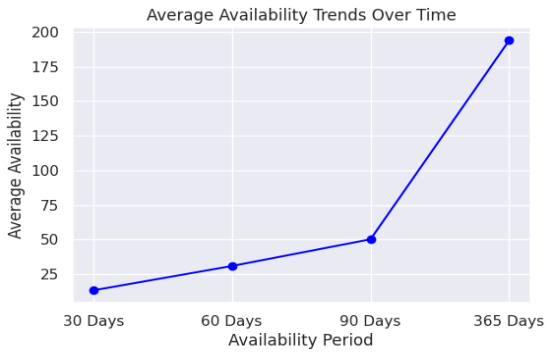




Figure 19.1: Average Availability, Figure 19.2: Price Vs. Availability

The graph is clearly demonstrating a positive relationship between the duration of the availability period and listings' average availability.

This plot indicates a positive relationship between the number of days available (over a specified time frame) and the listing price. The higher the number of days available, particularly when examining a longer time frame such as 365 days, the higher the average price and the wider range of prices.

5.4 INFERENTIAL STATISTICS

Inferential statistical techniques were used to find important relationships and differences among variables related to Airbnb listing prices. These analyses supported the patterns seen during exploratory data analysis by applying hypothesis testing frameworks.

5.4.1 Group Differences in Price by Room Type

A first one-way ANOVA was employed to compare differences in log-transformed price between room types. Although the ANOVA was very significant ($F = 709.80$, $p < 0.001$), its assumptions were breached:

- Shapiro-Wilk Test for Normality of Residuals: $p < 0.001$ — residuals are not normally distributed.
- Levene's Test for Homogeneity of Variance: $p < 0.001$ — variances are unequal across room types.

Due to these violations, ANOVA results are not entirely trustworthy. Therefore, two strong alternatives were used:

- Robust OLS Regression: Ordinary Least Squares regression was performed with HC3 robust standard errors to account for heteroscedasticity. All room type coefficients were statistically significant ($p < 0.001$), confirming meaningful differences in log-prices across categories.
- Kruskal-Wallis H-Test: A non-parametric alternative to ANOVA that does not require normality or equal variances. The test yielded $H = 1867.34$, $p < 0.001$, indicating statistically significant differences in median prices across room types.

These robust methods validate that room type significantly influences listing prices, even when classical ANOVA assumptions are not satisfied.

5.4.2 Effect of Host Super host Status on Price

A Welch's t-test was used to compare log-transformed prices between Super hosts and non-Super hosts, considering unequal variances. The results showed:

- Mean price (Super host): 5.0609
- Mean price (non-Super host): 5.0667
- t-statistic = -0.9675, p-value = 0.3333

Conclusion: There is no significant difference in mean price between Super hosts and non-Super hosts.

5.4.3 Association Tests (Chi-Square)

To find associations among categorical variables, a Chi-square test for independence was used:

- Super host Status vs. Instant Bookable: $\chi^2 = 15.65$, $p < 0.001$ — significant association; Super hosts have instant booking more frequently enabled.

5.4.4 Differences in Review Metrics

Kruskal-Wallis tests offered the analysis of differences for review-related variables by category groupings:

- Review Score Accuracy by Property Type: $H = 235.79$, $p < 0.001$ — accuracy differences in reviews are significant.
- Scores Value by Neighbourhood: $H = 270.20$, $p < 0.001$ — neighbourhoods are significantly different.

Overall, these inferential statistics validated and measured most of the trends observed in exploratory analysis. Use of robust as well as non-parametric techniques provided assurance against invalid results caused by violations of classical assumptions.

5.5 SPATIAL ANALYSIS

This section is devoted to clustering Airbnb listings based on their geographical location (latitude and longitude). The process was to use the K-Means algorithm for clustering in order to find the distinct clusters of listings based on their location. The images provided show the procedures and results of the clustering.

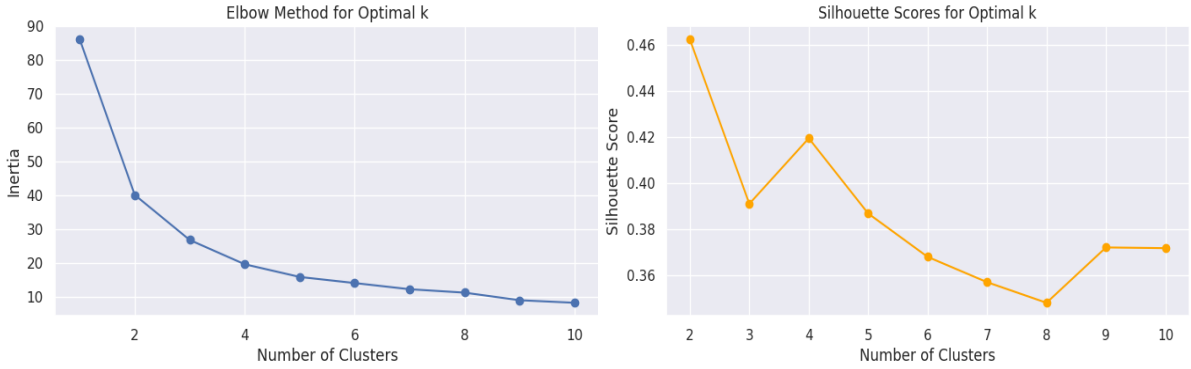


Figure 20.1: Elbow Method for Optimal k, Figure 20.2: Silhouette Scores for Optimal k

Elbow Method for Optimal k: This chart is utilized to determine the optimal value of the number of clusters (k) for the K-Means algorithm. The "elbow method" suggests finding the point at which the slope of the rate of decrease in inertia significantly changes, producing an "elbow" pattern. On this chart, the elbow appears to be at k=2 or k=3. Past this point, increasing the number of clusters produces diminishing returns in inertia decrease.

Silhouette Scores for Best k: This plot offers a second measure to assess the quality of clustering for various numbers of clusters and to assist in identifying the best number of clusters. The highest silhouette score in this plot occurs at k=2, indicating that the listings should be split into two clusters to obtain the most separated and cohesive clusters based on this measure.

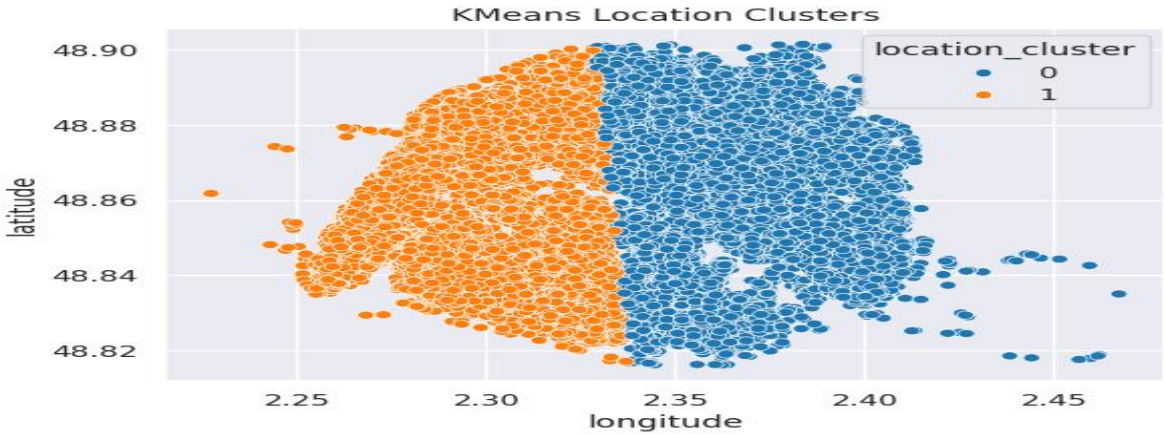


Figure 21: Location Cluster

K-Means Location Clusters: This is the output of running the K-Means algorithm with the optimal number of clusters determined (k=2 by silhouette score) on the latitude and longitude coordinates of the Airbnb listings. The plot clearly indicates the geographic division of the

Airbnb listings into two groups. The first one is concentrated in the western and central regions of the area presented, and the other is more visible in the eastern regions.

5.6 CORRELATION ANALYSIS

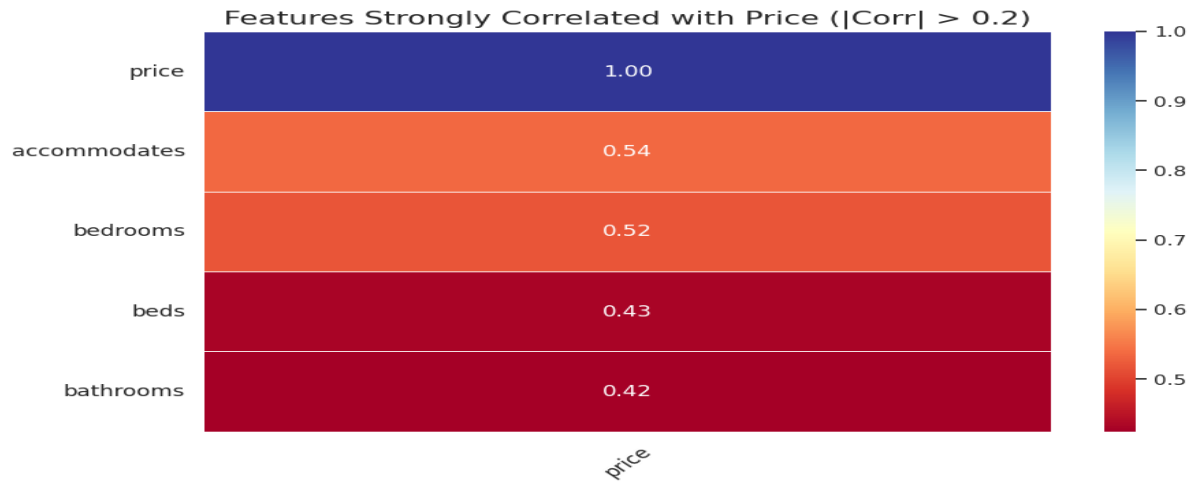


Figure 22: Correlation heatmap

This heatmap shows the numerical features with the strongest linear relationship to the listing price. The capacity for guests and the number of bedrooms has the strongest positive correlation with price among the features displayed. The number of beds and bathrooms also have a positive correlation, although it is a bit weaker.

Several Pearson correlation tests were done to assess linear relationships between continuous variables:

- Price vs. Accommodates: $r = 0.5474$, $p < 0.001$ — strong positive linear relationship.
- Number of Reviews vs. Review Scores Rating: $r = -0.0146$, $p < 0.001$ — very weak but statistically significant negative correlation.
- Host Response Rate vs. Host Acceptance Rate: $r = 0.4109$, $p < 0.001$ — moderate positive correlation.
- Minimum Nights vs. Availability (30 days): $r = -0.0341$, $p < 0.001$ — weak but statistically significant negative relationship.

These correlations reveal how various host and listing characteristics relate to price and availability.

CHAPTER 6
PREDICTIVE MODELING

6.1 DATA SPLITTING AND SCALING

In order to ready the data for machine learning model training and testing, the dataset was divided into training and testing sets. Feature scaling was then applied to ensure that the algorithms ran with optimal performance.

6.1.1 DATA SPLITTING

The data was split into a 80% training set and a 20% testing set by employing the train_test_split function from the sklearn.model_selection module. This stratification facilitates model training on the majority of the data while keeping a distinct, unseen portion for fair performance assessment. The features were isolated from the target attribute, price ('price'). An integer random seed (42) was employed in order to make the data split reproducible across different runs.

6.1.2 FEATURE SCALING

Numerous machine learning models are sensitive to the scaling of the input features. In order to reduce this and make sure that all features have an equal contribution to the learning process, feature scaling was performed using the StandardScaler of the sklearn.preprocessing module. Standardization converts every feature to have a mean of 0 and a standard deviation of 1. The scaler was trained only on the training data, and the parameters that were learned (mean and standard deviation) were applied to transform the training and test sets. This maintains the test data at the same scale as learned from the training data, mimicking the process of applying a trained model to novel, unseen data. The scaled training and testing features so obtained were applied in the subsequent steps of modelling.

6.2 DIMENSIONALITY REDUCTION: PCA

Principal Component Analysis (PCA) was employed to address the high dimensionality of our dataset, which contained a significant number of potentially correlated and redundant features. High dimensionality may result in the curse of dimensionality, longer training times, and difficulty in interpreting models. PCA achieves dimensionality reduction through the transformation of the original variables to a lower-dimension set of uncorrelated variables known as Principal Components (PCs). PCs are created to retain the most variance in the data, in effect compressing the original features' information content into a lower-dimensional space with minimal information loss. This dimensionality reduction seeks to enhance model performance, lower computational complexity, and simplify interpretability by capturing the important underlying patterns in the data.

6.2.1 NUMBER OF COMPONENTS RETAINED

Optimal principal components were decided by keeping sufficient components to explain a major percentage of the total variance. We wished to retain at least 95% of the total variance. From the explained variance analysis, 43 principal components were kept as together they retain at least 95% of the variance in the scaled training data. The fig represents the 95% variance threshold.

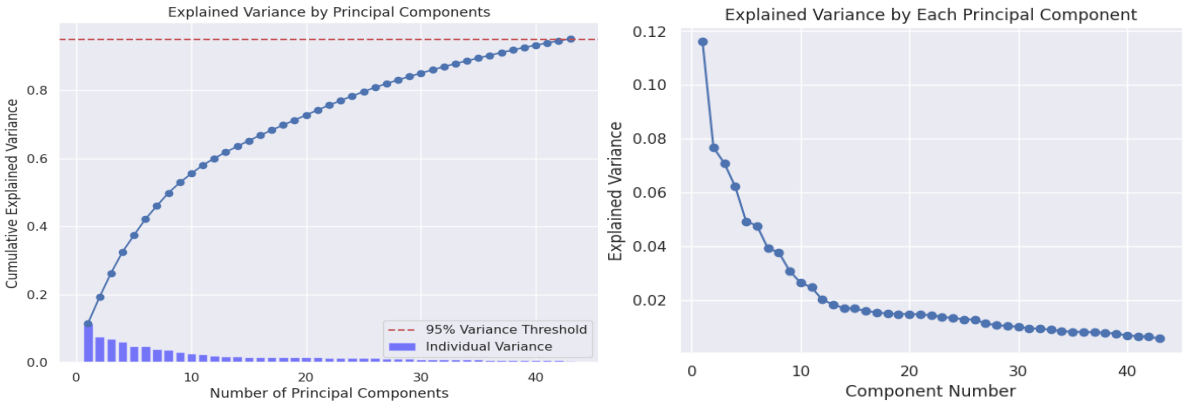


Figure 23.1: Cumulative Explained Variance by PC,
Figure 23.2: Explained Variance by Each PC

6.2.2 EXPLAINED VARIANCE

After PCA, explained variance was computed. Cumulative variance for the first 43 components ensures that about 95% of the original variance is retained in the transformed data set. The fig is a plot of explained variance by each principal component.

6.2.3 LOADING ANALYSIS HIGHLIGHTS

Principal component loadings indicate the contribution of each original feature to each principal component. The top contributing features for each principal component were examined to provide insight into the meaning of the principal components. This analysis reveals the original features with the strongest influence on each PC.

6.2.4 PCA-TRANSFORMED DATASETS

The PCA transformation created new training and test sets, with 43 columns for the principal components that were retained. The same parameters were used to transform the testing data as the training data, and it was transformed to keep results in line. These transformed data sets, with uncorrelated features preserving the most influential variations of the original data, were then utilized as input for other machine learning models.

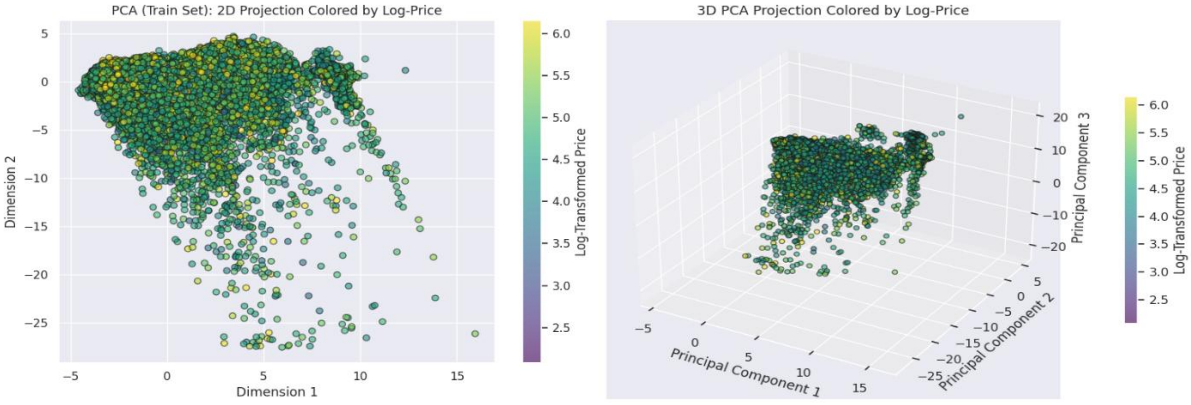


Figure 24.1: 2D Projection of PC, Figure 24.2: 3D Projection of PC

The plots, "PCA (Train Set): 2D Projection Coloured by Log-Price" and "3D PCA Projection Coloured by Log-Price," show the training data on the first two and three principal components, respectively, coloured by log-price. The plots visually validate that the principal components

are picking up variation in the data that aligns with the target variable (log-price), indicating their suitability for price prediction.

6.3 FEATURE SELECTION

Feature selection is an important process in machine learning for selecting and keeping the most important features that will be used to predict the target variable (Airbnb price), while eliminating the irrelevant or redundant features. Two main feature selection techniques were utilized in this research:

6.3.1 RANDOM FOREST FEATURE IMPORTANCE

A Random Forest Regressor was run on the dataset to determine feature importance. Random Forests naturally measure the contribution of every feature by how much the reduction in impurity (or error) is when splitting nodes on that feature. Features that result in more error reduction are considered more important. This approach gives a ranking of features in terms of how predictive they are in the Random Forest model.

6.3.2 SELECTKBEST WITH F_REGRESSION

This technique uses a statistical means of feature selection. SelectKBest chooses the best 'k' features according to a scoring function. The f_regression scoring function was utilized to measure the linear dependency between every feature and the target variable. Features with larger F-statistics are deemed more statistically significant predictors of the target variable.

6.3.3 DETERMINING THE NUMBER OF SELECTED FEATURES

For the SelectKBest method, the best number of features ('k') was established with cross-validation. The values of 'k' were iterated over, and for each 'k', the top 'k' features were chosen, a Random Forest Regressor was trained and assessed with cross-validation. Negative Mean Squared Error (Negative MSE) was employed as the performance measure. The 'k' value that gave the optimal average Negative MSE over cross-validation folds was selected. The Random Forest Feature Importance approach gave an importance ranking of features so that a subset of top-ranking features could be selected based on their importance scores.

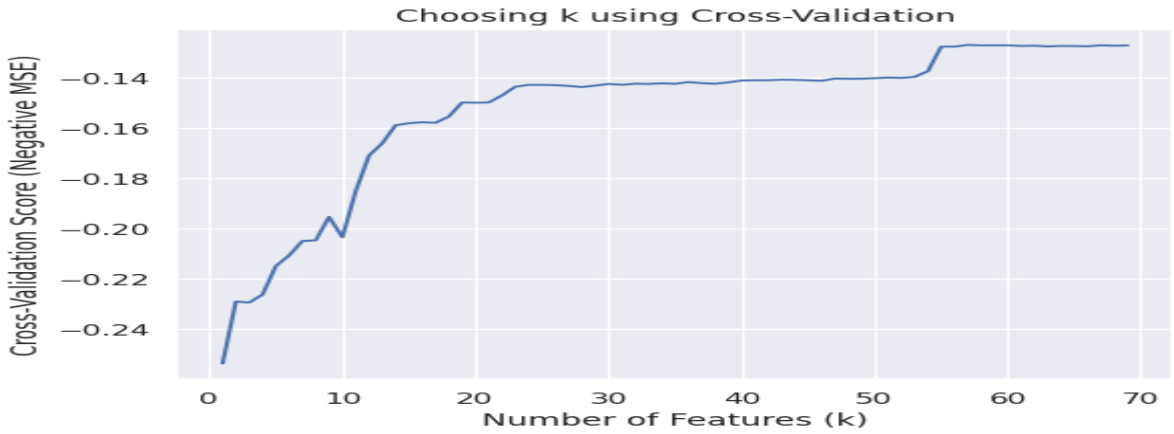


Figure 25: Cross Validation Plot

6.3.4 SELECTED FEATURES

From the cross-validation results using SelectKBest and the Random Forest Feature Importance, a subset of features was chosen. SelectKBest predicted the number of best features, and the chosen features were those with the top F-statistics. The Random Forest feature importance analysis also gave insight into the ability of these chosen features to predict in the Random Forest model.

6.3.5 FEATURE-SELECTED DATASETS

After feature selection, the initial training (x_train) and test (x_test) datasets were converted to contain just the chosen features, thereby yielding the feature-selected training dataset (x_train_selected) and the feature-selected test dataset (x_test_selected). These datasets, with only the most important features identified as relevant for Airbnb price prediction, were employed for subsequent model training and assessment.

6.4 MODEL TRAINING AND EVALUATION (FEATURE-SELECTED DATA)

The following regression models were trained using the feature-selected training data:

- Linear Regression
- Random Forest Regressor
- Decision Tree Regressor
- XGBoost Regressor
- LightGBM Regressor

Training for each model included fitting the model on the training data and making predictions on the test set. The target variable ('price') was log-transformed during data preprocessing, the prediction values were converted back to the original scale using the inverse transformation (e.g., np.expm1 for the natural logarithm) before computing the evaluation metrics.

Model performance was evaluated using the following metrics:

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R-squared (R^2)
- Mean Absolute Error (MAE)
- Mean Absolute Percentage Error (MAPE)

Table 2: Model Performance Table (Feature Selection)

Model	MSE	RMSE	R^2	MAE	MAPE
Linear Regression	7687.83	87.68	0.4817	58.60	32.02%
Random Forest	7599.41	87.17	0.4877	60.87	34.75%
Decision Tree Regressor	7791.34	88.27	0.4747	61.83	35.45%
XGBoost	4702.54	68.58	0.6830	45.07	24.81%
LightGBM	4719.37	68.70	0.6818	45.26	24.78%

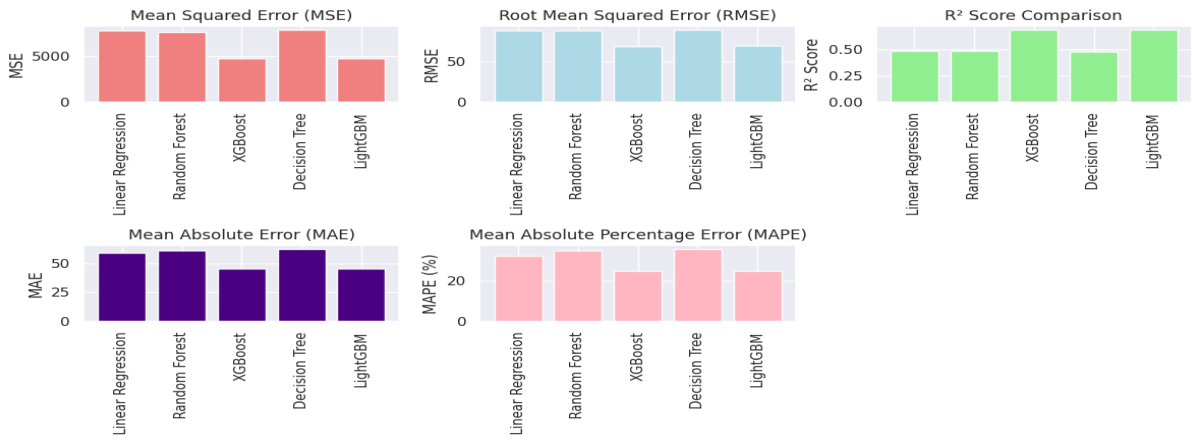


Figure 26: Model Performance

Hyperparameter tuning

Hyperparameter tuning was done for XGBoost and LightGBM with RandomizedSearchCV.

Table 3: Tuned Model Performance

Model	MSE	RMSE	R^2	MAE	MAPE
Tuned XGBoost	4554.52	67.49	0.6929	43.71	24.09%
Tuned LightGBM	4453.70	66.74	0.6997	43.52	23.79%

Visualizations:

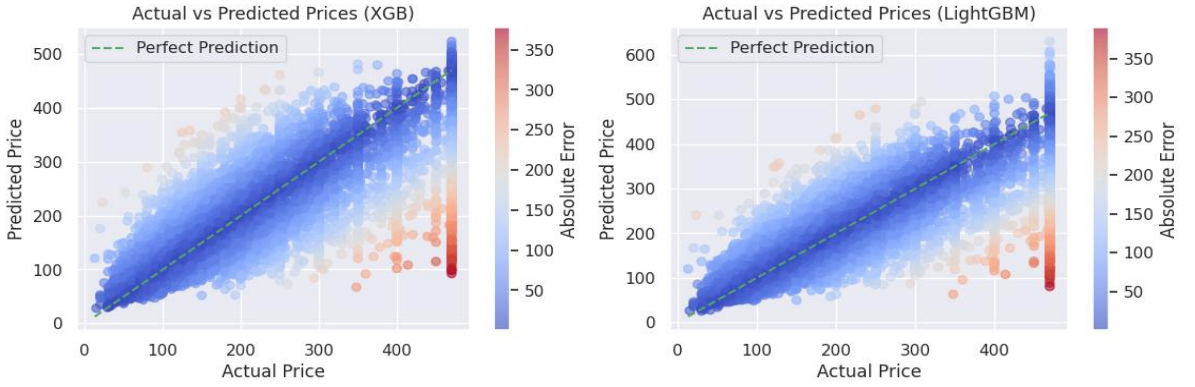


Figure 27.1: Actual vs Predicted of XGBoost, Figure 27.2: Actual vs Predicted of LightGBM

6.5 MODEL INTERPRETATION

Two gradient boosting algorithms, XGBoost and LightGBM, were used to test the importance of various features in the dataset.

- Geographic Location: Both models show that longitude is an extremely significant feature. The LightGBM model also suggests that latitude is a strong predictor. This shows the very strong effect of the geographical location of the listings.
- Property Characteristics: The number of bathrooms and bedrooms are significant features, as seen by the XGBoost model.

- •Host Activity: The calculated_host_listings_count is recognized as important by the XGBoost model, indicating that the scope of a host's listings is an important consideration.
- •Review Metrics: The XGBoost model focuses on the significance of review_scores_rating, whereas the LightGBM model places importance on reviews_per_month. This highlights the influence of guest reviews.
- •Availability and Amenities: The LightGBM model considers availability_365 and number_of_amenities to be significant, reflecting the importance of being available throughout the year and the variety of amenities provided.

The study indicates that the site of the property, its square footage (number of bathrooms and bedrooms), the activity level of the host, guest feedback, the property's yearly calendar availability, and the number of amenities offered are all highly predictive of the target variable. These relationships bear further investigation, which may provide useful insights.

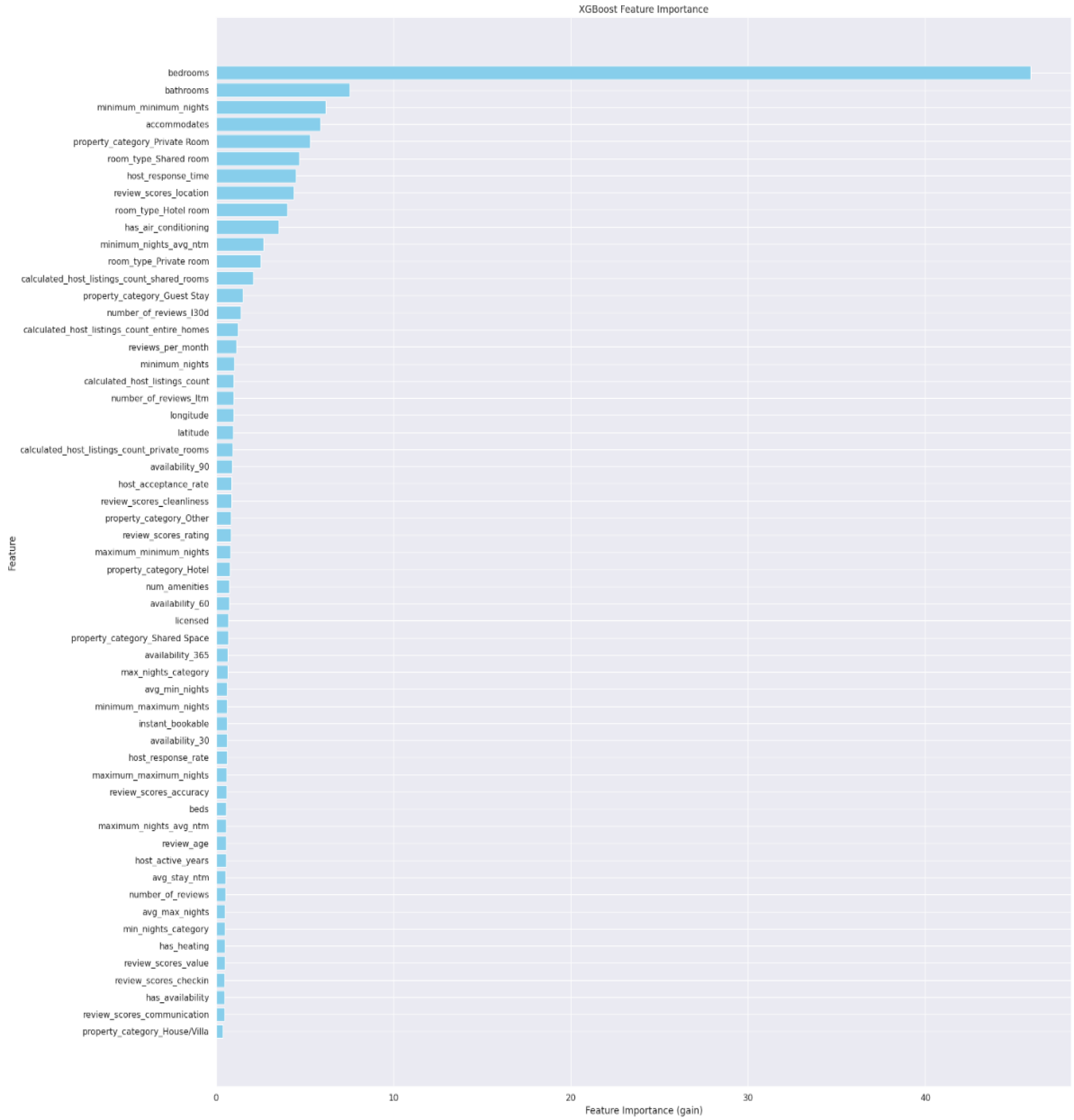


Figure 28: Feature Importance of XGBoost

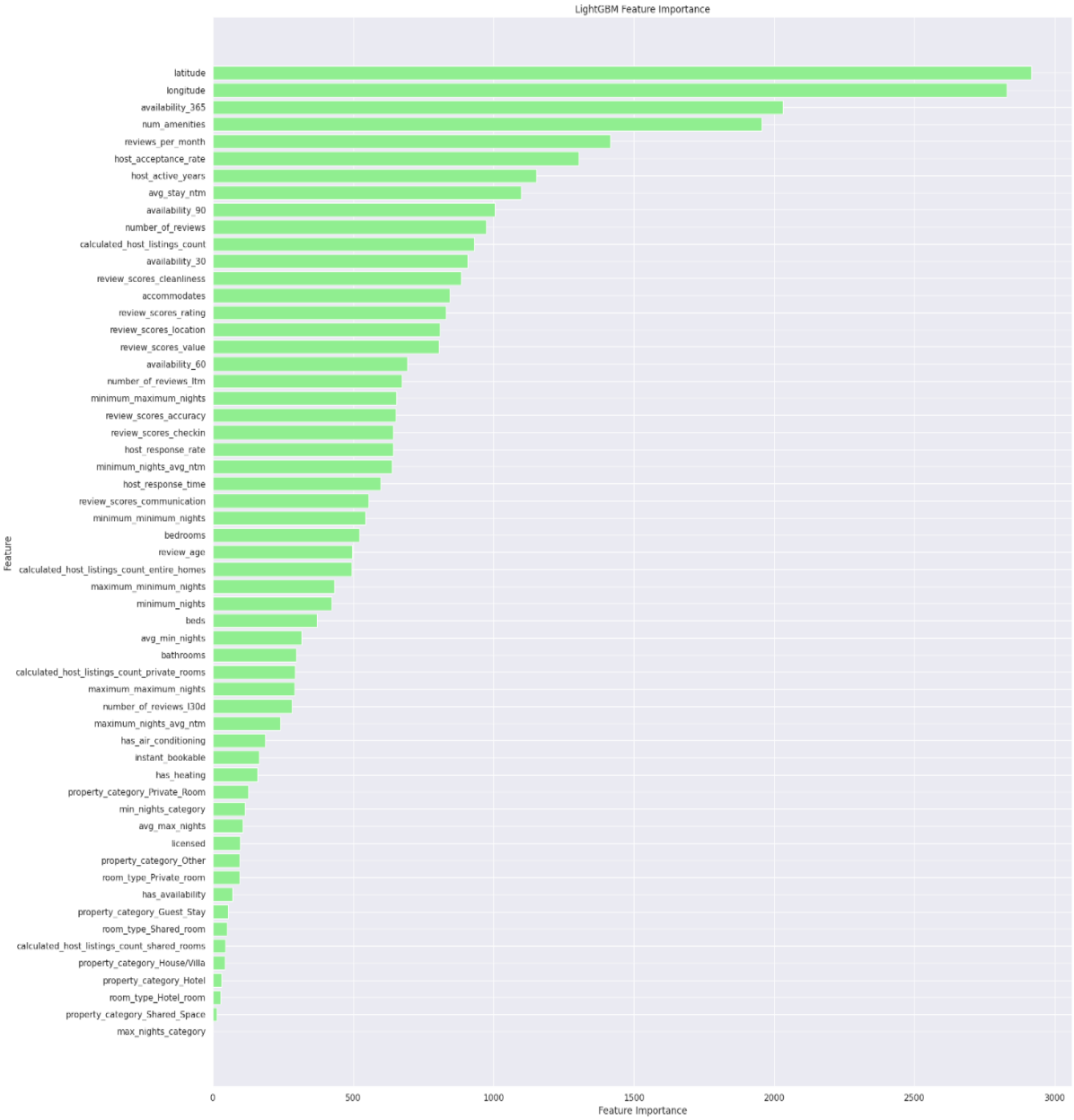


Figure 29: Feature Importance of LightGBM

6.6 STATISTICAL COMPARISON OF BEST MODELS

In order to further compare the performance of the two top-performing models after hyperparameter tuning (Tuned XGBoost and Tuned LightGBM), the difference between their predictions was checked using a statistical test. A two-sample independent t-test (scipy.stats.ttest_ind) was employed to compare the mean predictions of the tuned XGBoost model (xgb_preds) and the tuned LightGBM model (lgb_preds on the test set).
This test helps assess whether the observed differences in their performance metrics are statistically significant or could have occurred by chance.
The results of the t-test were:

- T-statistic: 0.24598183322579453
- P-value: 0.8056984649976021
- Fail to reject the null hypothesis: There is no statistically significant difference between predictions from XGBoost and LightGBM

As seen from the p-value (0.8056984649976021), which is higher than the 0.05 significance level, we cannot reject the null hypothesis. This implies that there is no statistically significant difference between the mean performance of tuned XGBoost and LightGBM models on the test set. Although their performance measures are slightly different, the differences are not statistically significant at the 0.05 level.

6.7 PCA-BASED MODELLING (Alternative Approach)

The PCA-transformed training data were also used to train the regression models. Nevertheless, because of low accuracy, feature selection was considered a better method here.

Table 4: Model Performance Table (PCA)

Model	MSE	RMSE	R^2	MAE	MAPE
Linear Regression	8046.34	89.70	0.4575	60.03	32.85%
Random Forest	8692.61	93.23	0.4140	65.41	37.67%
Decision Tree Regressor	9101.27	95.40	0.3864	67.35	39.11%
XGBoost	5775.33	76.00	0.6106	50.78	28.48%
LightGBM	5799.37	76.15	0.6090	50.97	28.38%

6.8 INTERACTIVE PRICE PREDICTION TOOL

To enable user interaction with the trained models and enable an intuitive grasp of price predictions, an interactive interface was created using the ipywidgets library in the notebook environment. This interface enables users to directly enter values for the most prominent features determined in feature selection and see the corresponding predicted listing price in real-time from the top-performing models (tuned XGBoost and LightGBM).

Implementation:

For every selected_feature, a suitable ipywidget was created according to the type of data of the feature: widgets.IntText for integers, widgets.FloatText for floats (both created with the mean of the feature), and widgets.BoundedIntText for presumably binary features (created with the mode and bounded between 0 and 1). These widgets were named according to their respective feature names and ordered for user input.

A specialized widgets.Output() space (output_widget) was established in which to present the prediction output, keeping the output tidy and sharp. The main functionality is contained within the predict_price function. This function runs automatically whenever a user changes the value of any input widget. It does the following:

1. Input Acquisition: Grabs the current values of all input widgets and saves them in a dictionary.

2. Data Formatting: Transforms the input dictionary into a Pandas DataFrame with a single row, the desired input format of the prediction models.

3. Prediction Generation: Employ the trained best XGBoost model (model_xgb_best) and the best LightGBM model (best_lgb) to make price predictions from the user input.

4. Inverse Transformation: Uses the inverse of the logarithmic transformation (np.expm1) on the predicted price for reversal back to its original scale to facilitate interpretability.

5.Result Display: Clears out any existing output within the output_widget and then outputs the newly calculated price, rounded to two decimal places.

Dynamic updating of the prediction is facilitated through the attachment of an observer to the value property of every input widget via the widget.observe(predict_price, names='value') method. This calls the predict_price function automatically whenever an input value is modified.

Lastly, the input_widgets and the output_widget are shown on the notebook through the display () function, creating a friendly interactive tool.

This interactive platform is a useful means of examining the behaviour of the trained models. In enabling users to manipulate the input features that the models themselves have recognized as most impactful, it helps to gain a more intuitive appreciation of how these components determine the predicted listing price. This can be useful for stakeholders in obtaining insight into the model's thought process and in conducting "what-if" analysis by simulating various scenarios.

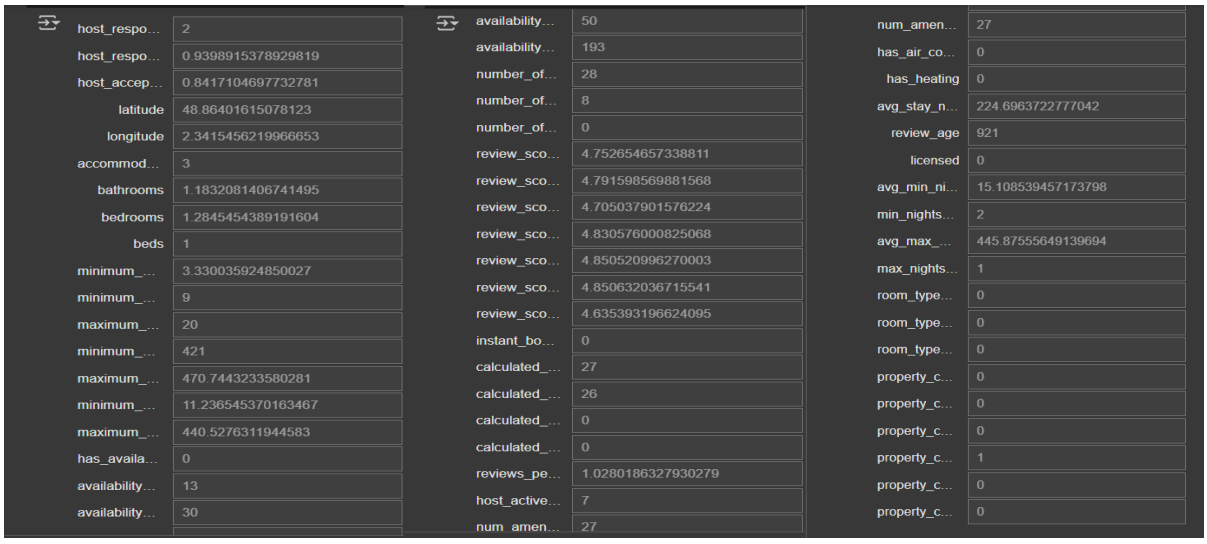


Figure 30: Interactive Model

6.9 DISCUSSION AND FINDINGS

- Preprocessing (scaling, log transformation) greatly improved performance.
- Feature selection worked better than PCA on model performance.
- Tree-based models, particularly tuned XGBoost and LightGBM, gave the best predictive accuracy.
- Geographic location, size of the property, host activity, availability, and amenities were high predictive features.
- No difference of statistical significance between the top two models' predictions.
- The interactive tool gave interpretability and application to the model.

This chapter showed the full machine learning pipeline, from data preparation to interpretation, to make a successful price prediction for Airbnb listings in Paris.

CHAPTER 7
CONCLUSIONS AND RECOMMENDATIONS

7.1 SUMMARY OF KEY FINDINGS

The project made a thorough data-driven attempt to forecast the prices of Airbnb listings in Paris. The key findings of each phase are encapsulated below:

- Data Preparation: Heavily cleaning and preprocessing were carried out, including the removal of outliers, log transformation of skewed prices, encoding categorical variables, and sentiment analysis of reviews. Sentiment analysis was restricted because of French-language reviews.
- Exploratory Analysis: Univariate and multivariate analyses revealed some useful information regarding the distribution and patterns of features that impact price, including location, room type, and host activity.
- Inferential Statistics: Statistical tests confirmed the significance of price differences observed among room types, property types, and locations. Kruskal-Wallis and Welch's t-tests were applied as there were violations of normality and equal variance assumptions. Robust OLS regression and Pearson correlation also confirmed such relationships, for example, the positive linear relationship between accommodates and price.
- Feature Engineering & Selection: Dimensionality reduction and model efficiency were improved using PCA and statistical feature selection techniques.
- Modeling & Evaluation: Multiple regression models were trained, and XGBoost and LightGBM proved to be the best. The performance of these models was enhanced further by hyperparameter tuning.
- Interactive Prediction Interface: An interface based on ipywidgets was developed so that users could make simulated predictions and obtain interpretability.

7.2 RESEARCH CONTRIBUTIONS

- Implemented an end-to-end data science workflow from data gathering to deployment.
- Confirmed the significance of host, in-time, and space variables in determining price.
- Developed interpretable, high-quality predictive models for Airbnb listings.
- Offered an interactive price simulation tool with respect to input features.

7.3 LIMITATIONS

- Geographical Scope: The study was restricted to Paris. Results might not hold for other cities without retraining the model.
- Temporal Constraints: Calendar data only reflected a particular snapshot in time.
- Textual Reviews: Sentiment analysis was limited because the reviews were primarily in French, restricting the effectiveness of natural language processing. Deeper NLP methods or translation could offer richer insights.
- Platform Limitations: The current interface is limited to Google Colab and lacks broader web accessibility.

7.4 RECOMMENDATIONS

- For Hosts:

    - Focus on property features like full apartment offerings and high-quality amenities.
    - Maintain strong review scores and fast response rates to increase perceived value.
- For Airbnb Platform:
    - Integrate automated price suggestion tools using models like this research.
    - Promote hosts to enhance engagement and validate their listings.
- For Researchers:
    - Provide textual and visual attributes (e.g., reviews, images) for more comprehensive models.
    - Investigate causal effects or use time series forecasting methods.

7.5 FUTURE WORK

- Web-Based Interface: Implement the model as a web-based application with frameworks such as Flask or Streamlit.
- Multi-City Expansion: Expand the model to forecast prices in other top Airbnb cities.
- Deep NLP: Apply transformer-based models such as BERT for more profound sentiment and review analysis, particularly after review translation.
- Real-Time Updates: Incorporate streaming data to enable dynamic retraining and more precise predictions.

This research forms a sound basis for further studies and real-world implementation in actual Airbnb pricing applications. It bridges the gap between scholarly understanding and real-world applicability in the fast-changing sharing economy.

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